Tugas Pertemuan 7 - DML: Join Tables

Mata Kuliah: Sistem Basis Data (2026)

Prodi: Sarjana Terapan Teknik Informatika POLBAN

Kelas: 2B

NIM: 251524048

Nama Lengkap: Ghaisan Khoirul Badruzaman

Prasyarat

1. Menggunakan alat bantu atau perangkat lunak RDBMS PostgreSQL
2. Menggunakan GUI Client Tools seperti pgAdmin atau DBeaver (opsional)

Instruksi

1. Lakukan import database atau schema HR Oracle untuk PostgreSQL yang telah disediakan: https://www.modb.pro/db/100650
2. Praktikkan materi pada slide terlampir!
3. Praktikkan materi secara terurut!
4. Buatkan dokumentasi hasil praktik, harus mencakup
   - Judul Bagian yang dipraktikkan
   - Syntax Query
   - Keterangan (Jika ada penjelasan yang ingin ditambahkan, tambahkan di sini)
   - Screenshot Output (Sertakan screenshot halaman penuh yang menyertakan taskbar dengan waktu (Print Screen) untuk menunjukkan syntax dan hasil eksekusi query pada RDBMS!)
5. Dalam suatu bagian materi, Anda boleh mempraktikkan dengan beberapa jenis percobaan! Tuliskan judul bagian yang jelas/ spesifik!
6. Sertakan NIM, Nama Lengkap, Nama Mata Kuliah pada isi dokumen Anda!
7. Saat mengerjakan pasti akan mengalami kesulitan, karena contoh yang ada pada slide, merupakan syntax untuk RDBMS Oracle. Sehingga Anda harus menyesuaikan syntax yang dapat dieksekusi pada RDBMS **PostgreSQL**!

Deliverable & Deadline

1. Ekspor file lembar jawaban ini dalam format PDF dengan nama file:
   Tugas_P7_<2X - Kelas>_<3 Digit Terakhir NIM>.pdf
   contoh sebagai berikut:
   Tugas_P7_2A_003.pdf
2. Kumpulkan melalui Assignment Teams sesuai dengan tenggat waktu yang telah ditentukan!

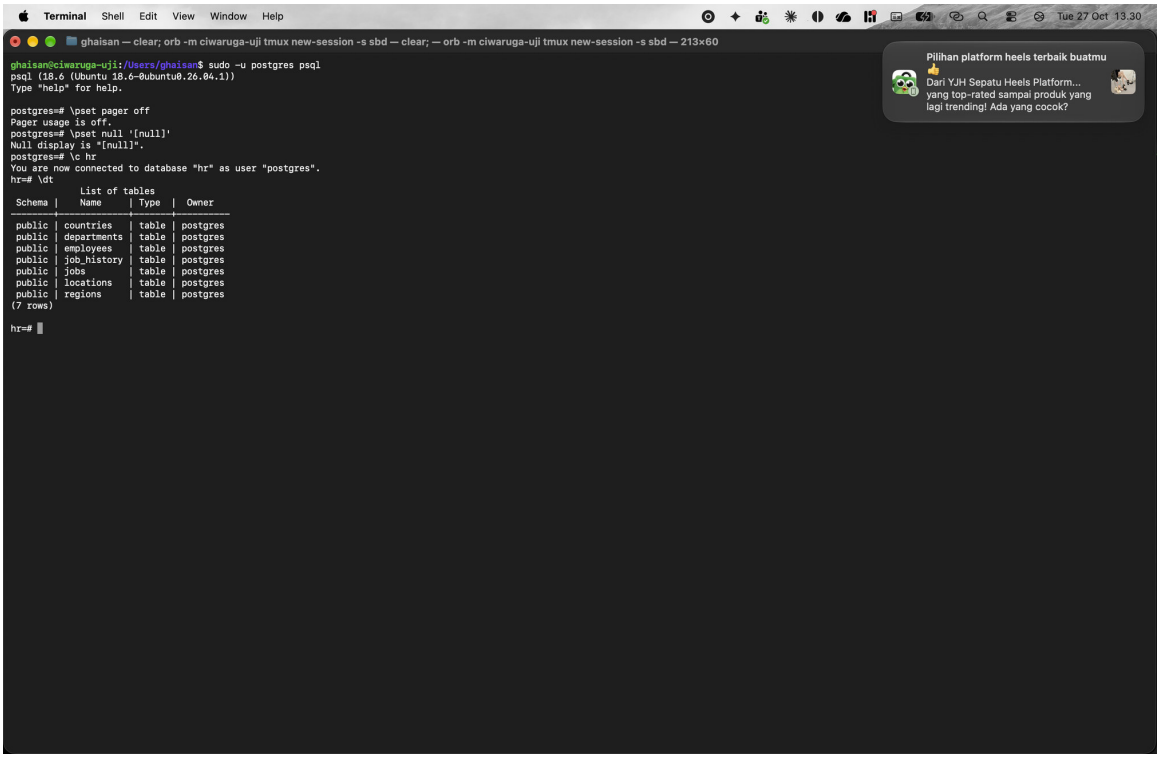
Jika ada hal yang ingin ditanyakan, silahkan sampaikan melalui Teams.

Selamat mengerjakan!

Latihan No. 1

Judul
Masuk ke database hr (schema HR Oracle untuk PostgreSQL dari modb.pro/db/100650)
Syntax

`sudo -u postgres psql`
 	
`\pset pager off`
 	
`\pset null '[null]'`
 	
`\c hr`
 	
`\dt`

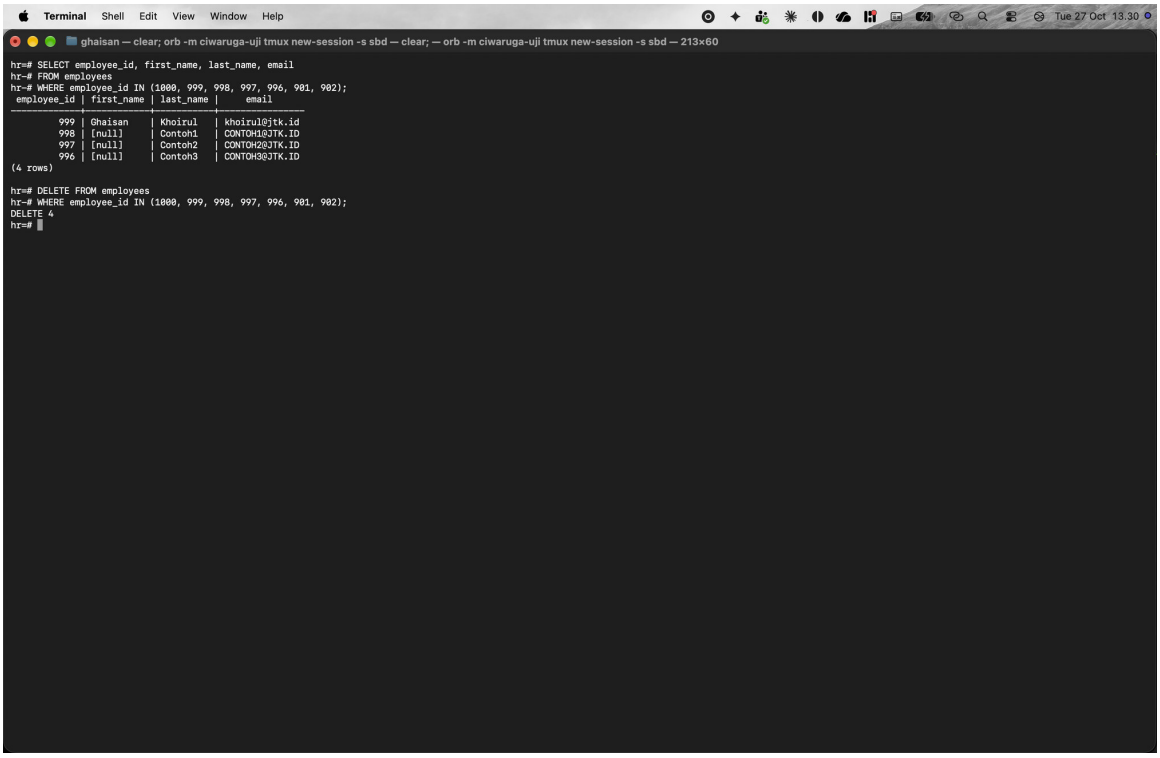
Keterangan
Schema HR untuk PostgreSQL dari modb.pro/db/100650 sudah diimport ke database hr pada Praktikum 6 (file hr_schema.sql dijalankan dengan \i), jadi di praktikum ini cukup masuk ke psql lalu pindah ke database hr dengan \c hr. \dt menampilkan 7 tabel hasil import. \pset pager off supaya hasil yang panjang langsung tampil tanpa pager, dan \pset null '[null]' supaya NULL kelihatan jelas, sama seperti tampilan [null] di pgAdmin pada slide.
Hasil Eksekusi (Screenshot)


Latihan No. 2

Judul
Number of Rows Per Table: menghapus data percobaan pertemuan sebelumnya
Syntax

`SELECT employee_id, first_name, last_name, email`
`FROM employees`
`WHERE employee_id IN (1000, 999, 998, 997, 996, 901, 902);`
 	
`DELETE FROM employees`
`WHERE employee_id IN (1000, 999, 998, 997, 996, 901, 902);`

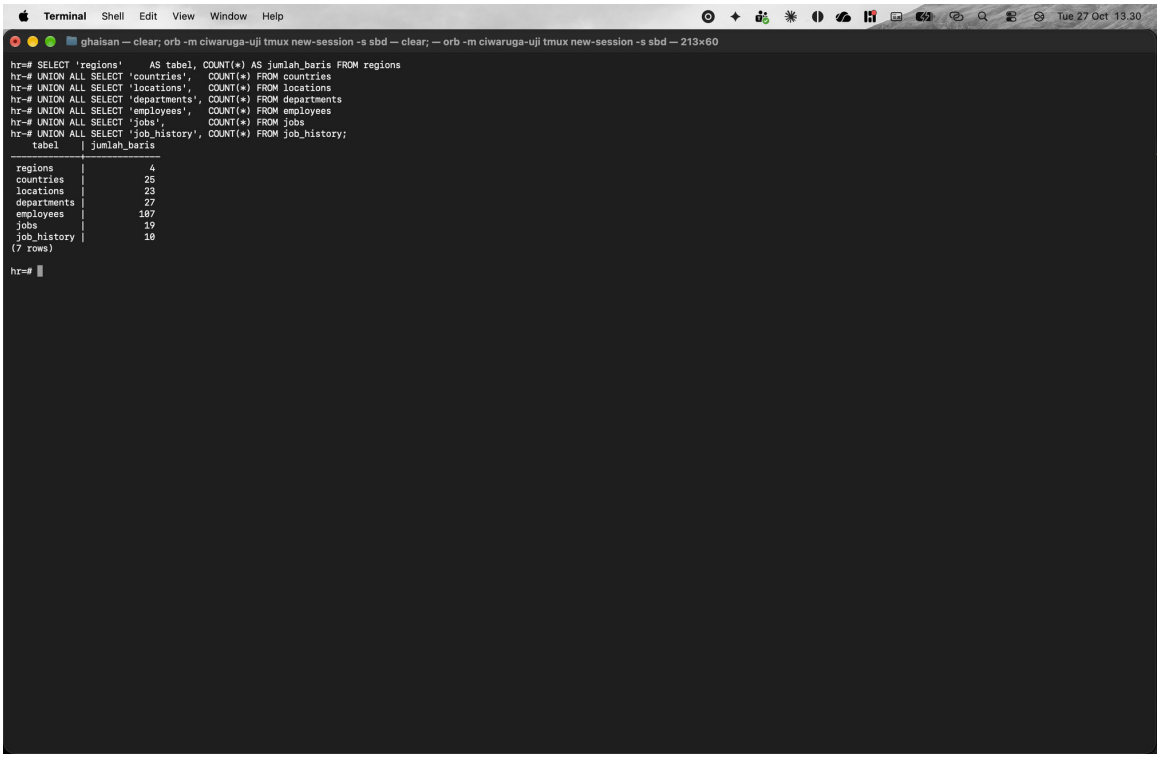
Keterangan
Slide meminta data percobaan pertemuan sebelumnya dihapus dulu supaya isi tabel kembali seperti data asli. SELECT dijalankan dulu untuk melihat baris yang akan dihapus: 4 pegawai (996 sampai 999) sisa latihan INSERT di Praktikum 6. Pegawai 1000 sudah dihapus di Praktikum 6, sedangkan 901 dan 902 tidak pernah dibuat, jadi hasilnya DELETE 4 (di slide DELETE 3 karena data percobaannya berbeda).
Hasil Eksekusi (Screenshot)



Latihan No. 3

Judul
Number of Rows Per Table: jumlah baris setiap tabel
Syntax

`SELECT 'regions'     AS tabel, COUNT(*) AS jumlah_baris FROM regions`
`UNION ALL SELECT 'countries',   COUNT(*) FROM countries`
`UNION ALL SELECT 'locations',   COUNT(*) FROM locations`
`UNION ALL SELECT 'departments', COUNT(*) FROM departments`
`UNION ALL SELECT 'employees',   COUNT(*) FROM employees`
`UNION ALL SELECT 'jobs',        COUNT(*) FROM jobs`
`UNION ALL SELECT 'job_history', COUNT(*) FROM job_history;`

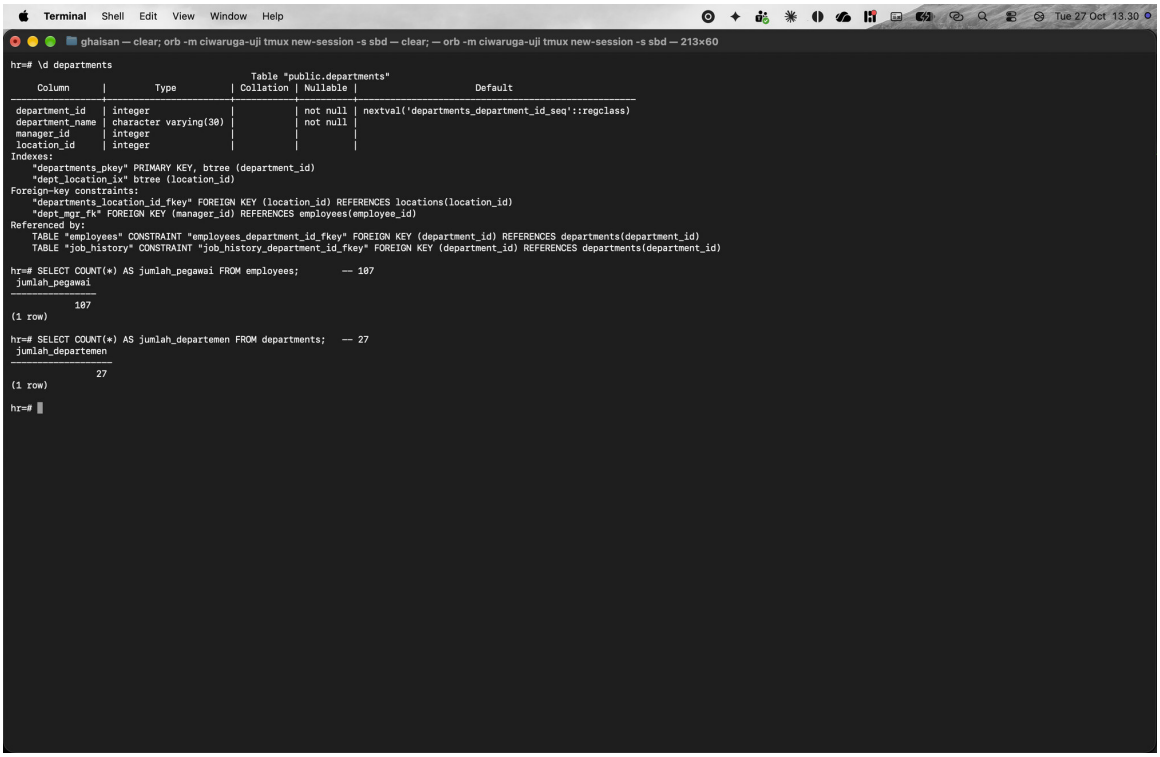
Keterangan
Jumlah baris tiap tabel sudah sama dengan slide: regions 4, countries 25, locations 23, departments 27, employees 107, jobs 19, dan job_history 10. UNION ALL dipakai supaya tujuh hasil COUNT ditumpuk jadi satu tabel. Operator UNION dibahas lagi di bagian akhir.
Hasil Eksekusi (Screenshot)


Latihan No. 4

Judul
Employees and Departments Tables: kolom penghubung department_id dan jumlah baris
Syntax

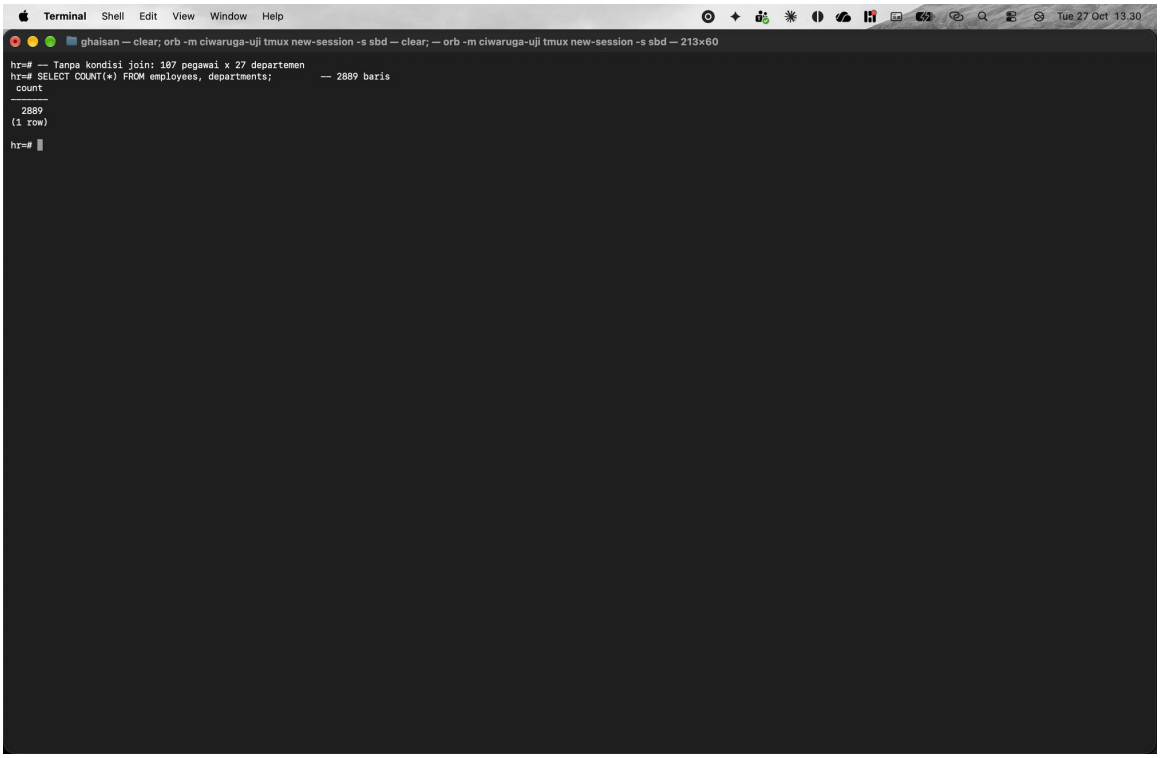
`\d departments`
 	
`SELECT COUNT(*) AS jumlah_pegawai FROM employees;        -- 107`
 	
`SELECT COUNT(*) AS jumlah_departemen FROM departments;   -- 27`

Keterangan
\d departments menunjukkan department_id adalah primary key tabel departments (departments_pkey). Bagian Referenced by menunjukkan kolom department_id di tabel employees adalah foreign key (employees_department_id_fkey) yang mengacu ke departments. Kolom inilah yang dipakai sebagai kondisi join di latihan berikutnya. Jumlah barisnya 107 pegawai dan 27 departemen.
Hasil Eksekusi (Screenshot)



Latihan No. 5

Judul
Join dengan klausa WHERE: tanpa kondisi join (Cartesian product)
Syntax
`-- Tanpa kondisi join: 107 pegawai x 27 departemen` `SELECT COUNT(*) FROM employees, departments;  -- 2889 baris`
Keterangan
Kalau dua tabel ditulis di FROM tanpa kondisi join, setiap baris employees dipasangkan dengan setiap baris departments (Cartesian product), jadi hasilnya 107 x 27 = 2889 baris.

Hasil Eksekusi (Screenshot)


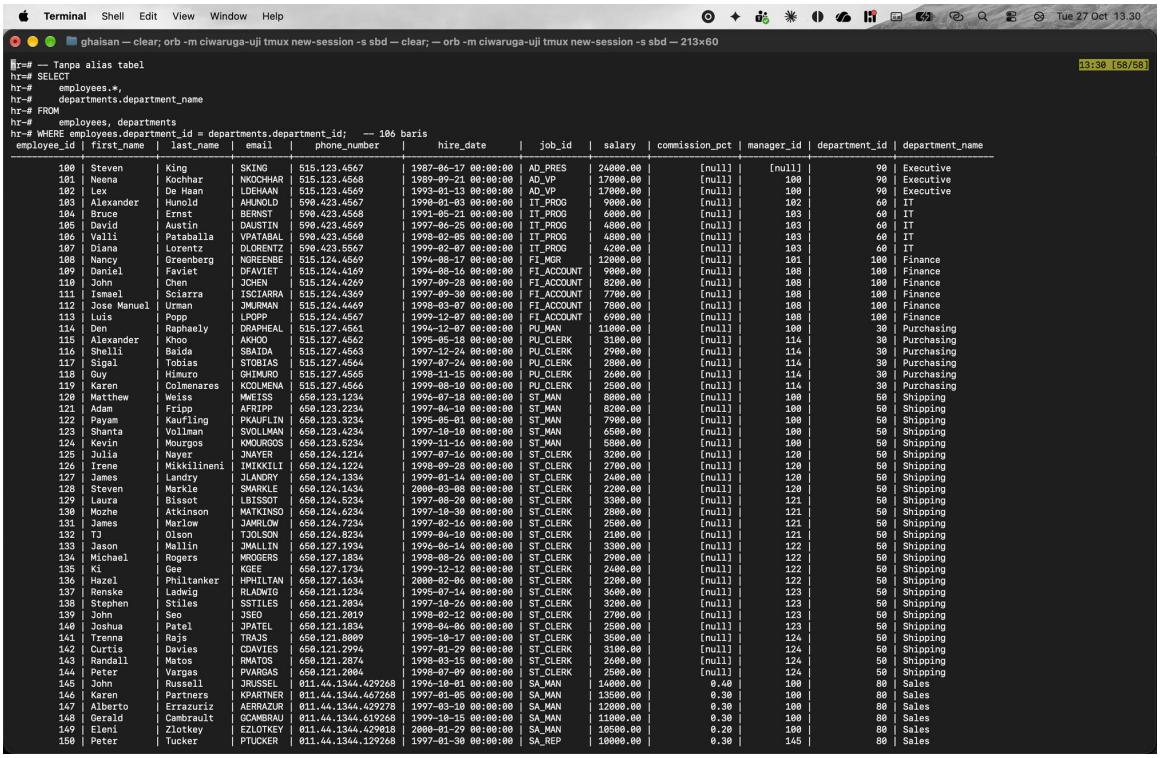
Latihan No. 6

Judul
Join dengan klausa WHERE: tanpa alias tabel
Syntax

`-- Tanpa alias tabel`
`SELECT`
`  employees.*,`
`  departments.department_name`
`FROM`
`  employees, departments`
`WHERE employees.department_id = departments.department_id;  -- 106 baris`

Keterangan
Kondisi employees.department_id = departments.department_id di WHERE menyaring pasangan baris yang cocok saja, hasilnya 106 baris. Tanpa alias, nama tabel harus ditulis lengkap di depan setiap kolom. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.30
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# — Tanpa alias tabel                                                                                  13:30 [58/58]
hr=# SELECT
hr=#     employees.*,
hr=#     departments.department_name
hr=# FROM
hr=#     employees, departments
hr=# WHERE employees.department_id = departments.department_id;   — 106 baris
 employee_id | first_name  | last_name   | email    |    phone_number    |      hire_date      |   job_id   |  salary  | commission_pct | manager_id | department_id | department_name
-------------+-------------+-------------+----------+--------------------+---------------------+------------+----------+----------------+------------+---------------+-----------------
         100 | Steven      | King        | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena       | Kochhar     | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex         | De Haan     | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander   | Hunold      | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce       | Ernst       | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David       | Austin      | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli       | Pataballa   | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana       | Lorentz     | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy       | Greenberg   | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel      | Faviet      | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John        | Chen        | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael      | Sciarra     | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman       | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis        | Popp        | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den         | Raphaely    | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander   | Khoo        | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli      | Baida       | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal       | Tobias      | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy         | Himuro      | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen       | Colmenares  | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew     | Weiss       | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam        | Fripp       | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam       | Kaufling    | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta      | Vollman     | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin       | Mourgos     | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia       | Nayer       | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene       | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James       | Landry      | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven      | Markle      | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura       | Bissot      | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe       | Atkinson    | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James       | Marlow      | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ          | Olson       | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason       | Mallin      | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael     | Rogers      | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki          | Gee         | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel       | Philtanker  | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske      | Ladwig      | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen     | Stiles      | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John        | Seo         | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua      | Patel       | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna      | Rajs        | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis      | Davies      | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall     | Matos       | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter       | Vargas      | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John        | Russell     | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen       | Partners    | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto     | Errazuriz   | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald      | Cambrault   | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni       | Zlotkey     | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter       | Tucker      | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
```

Latihan No. 7

Judul
Join dengan klausa WHERE: memakai alias tabel

Syntax

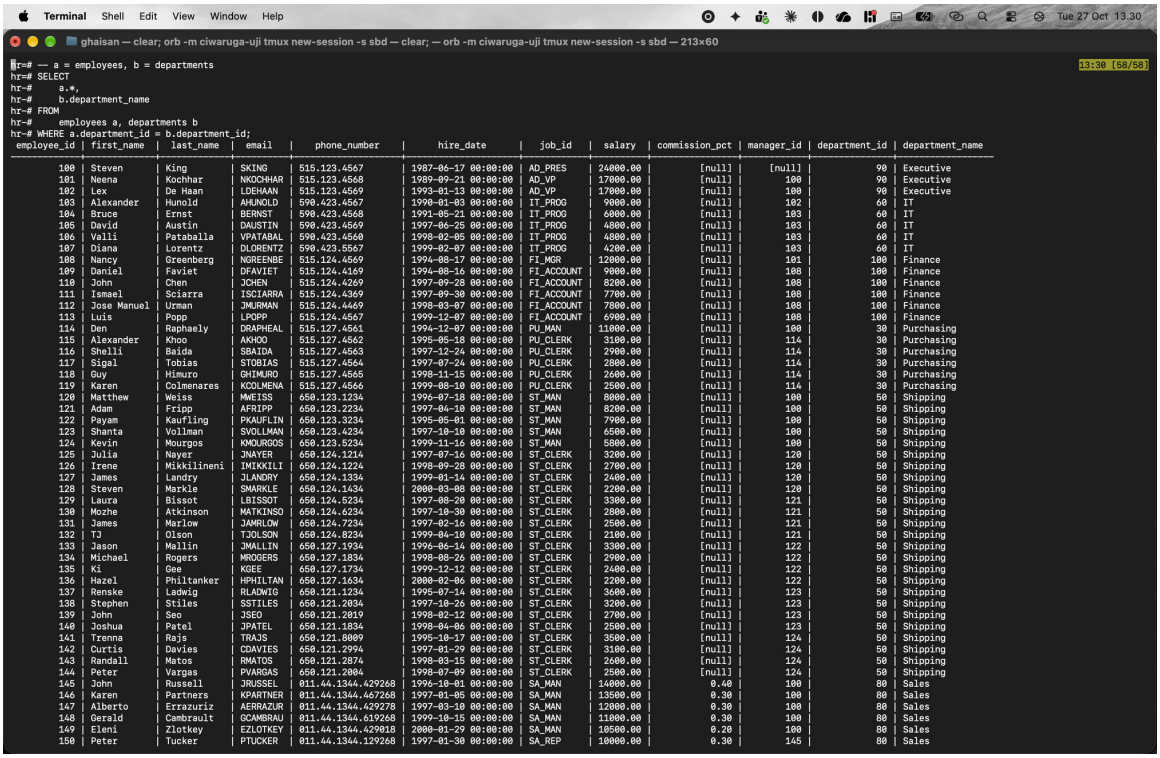
```
-- a = employees, b = departments
SELECT
    a.*,
    b.department_name
FROM
    employees a, departments b
WHERE a.department_id = b.department_id;
```

Keterangan

Hasilnya sama dengan latihan sebelumnya (106 baris). Penulisannya lebih ringkas karena employees diganti alias a dan departments diganti alias b.

Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.30
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# — a = employees, b = departments                                                                  13:30 [58/58]
hr=# SELECT
hr-#     a.*,
hr-#     b.department_name
hr-# FROM
hr-#     employees a, departments b
hr-# WHERE a.department_id = b.department_id;
 employee_id | first_name | last_name | email    | phone_number | hire_date           | job_id     | salary   | commission_pct | manager_id | department_id | department_name
-------------+------------+-----------+----------+--------------+---------------------+------------+----------+----------------+------------+---------------+-----------------
         100 | Steven     | King      | SKING    | 515.123.4567 | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena      | Kochhar   | NKOCHHAR | 515.123.4568 | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex        | De Haan   | LDEHAAN  | 515.123.4569 | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander  | Hunold    | AHUNOLD  | 590.423.4567 | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce      | Ernst     | BERNST   | 590.423.4568 | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David      | Austin    | DAUSTIN  | 590.423.4569 | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli      | Pataballa | VPATABAL | 590.423.4560 | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana      | Lorentz   | DLORENTZ | 590.423.5567 | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy      | Greenberg | NGREENBE | 515.124.4569 | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel     | Faviet    | DFAVIET  | 515.124.4169 | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John       | Chen      | JCHEN    | 515.124.4269 | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael     | Sciarra   | ISCIARRA | 515.124.4369 | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman    | JMURMAN  | 515.124.4469 | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis       | Popp      | LPOPP    | 515.124.4567 | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den        | Raphaely  | DRAPHEAL | 515.127.4561 | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander  | Khoo      | AKHOO    | 515.127.4562 | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli     | Baida     | SBAIDA   | 515.127.4563 | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal      | Tobias    | STOBIAS  | 515.127.4564 | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy        | Himuro    | GHIMURO  | 515.127.4565 | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566 | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew    | Weiss     | MWEISS   | 650.123.1234 | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam       | Fripp     | AFRIPP   | 650.123.2234 | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam      | Kaufling  | PKAUFLIN | 650.123.3234 | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta     | Vollman   | SVOLLMAN | 650.123.4234 | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin      | Mourgos   | KMOURGOS | 650.123.5234 | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia      | Nayer     | JNAYER   | 650.124.1214 | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224 | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James      | Landry    | JLANDRY  | 650.124.1334 | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven     | Markle    | SMARKLE  | 650.124.1434 | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura      | Bissot    | LBISSOT  | 650.124.5234 | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson  | MATKINSO | 650.124.6234 | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James      | Marlow    | JAMRLOW  | 650.124.7234 | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ         | Olson     | TJOLSON  | 650.124.8234 | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason      | Mallin    | JMALLIN  | 650.127.1934 | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael    | Rogers    | MROGERS  | 650.127.1834 | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki         | Gee       | KGEE     | 650.127.1734 | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634 | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske     | Ladwig    | RLADWIG  | 650.121.1234 | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen    | Stiles    | SSTILES  | 650.121.2034 | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John       | Seo       | JSEO     | 650.121.2019 | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua     | Patel     | JPATEL   | 650.121.1834 | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna     | Rajs      | TRAJS    | 650.121.8009 | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis     | Davies    | CDAVIES  | 650.121.2994 | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall    | Matos     | RMATOS   | 650.121.2874 | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter      | Vargas    | PVARGAS  | 650.121.2004 | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John       | Russell   | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen      | Partners  | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto    | Errazuriz | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald     | Cambrault | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni      | Zlotkey   | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter      | Tucker    | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
```

Latihan No. 8

Judul
Join dengan klausa WHERE: urutan tabel dibalik

Syntax

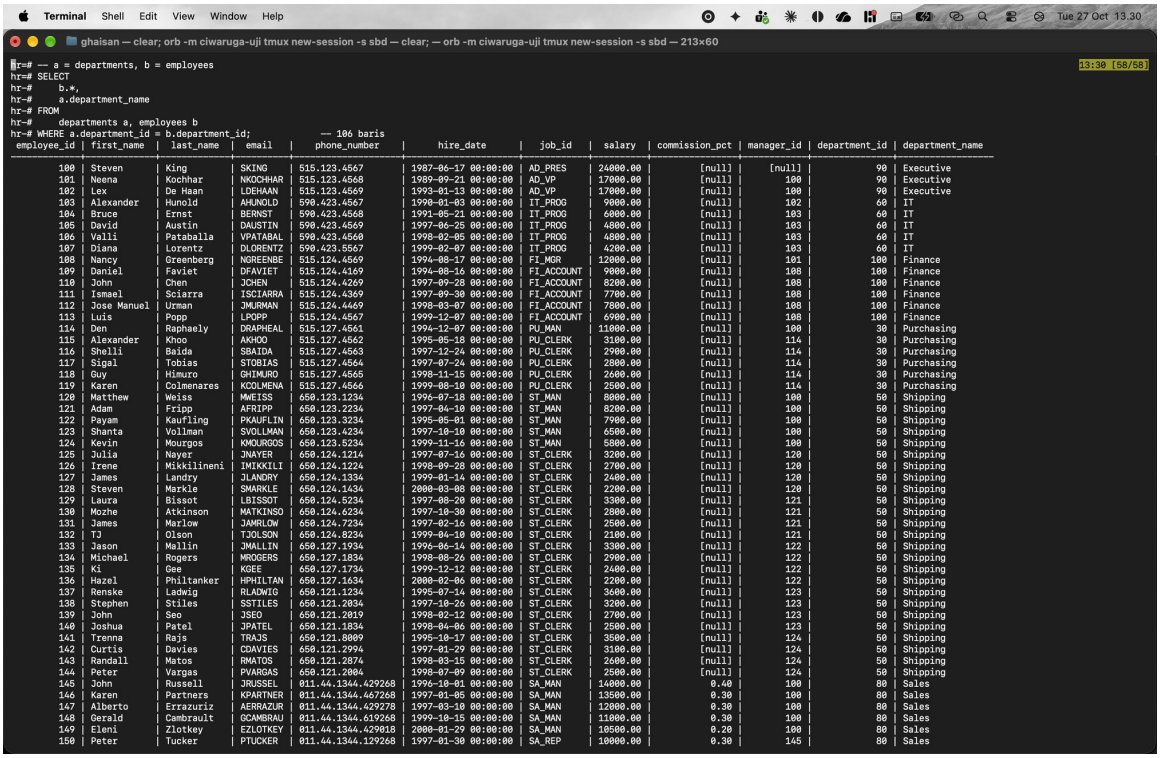
```
-- a = departments, b = employees
SELECT
    b.*,
    a.department_name
FROM
    departments a, employees b
WHERE a.department_id = b.department_id;              -- 106 baris
```

Keterangan

Urutan tabel di FROM tidak mengubah jumlah baris, tetap 106. Yang berubah hanya peran alias: sekarang a = departments dan b = employees, jadi b.* berisi kolom employees.

Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.30
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# -- a = departments, b = employees                                                      13:30 [58/58]
hr=# SELECT
hr-#     b.*,
hr-#     a.department_name
hr-# FROM
hr-#     departments a, employees b
hr-# WHERE a.department_id = b.department_id;          -- 106 baris
 employee_id | first_name | last_name | email  |    phone_number    |      hire_date      |   job_id   | salary  | commission_pct | manager_id | department_id | department_name
-------------+------------+-----------+--------+--------------------+---------------------+------------+---------+----------------+------------+---------------+-----------------
         100 | Steven     | King      | SKING  | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena      | Kochhar   | NKOCHHAR | 515.123.4568     | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex        | De Haan   | LDEHAAN | 515.123.4569      | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander  | Hunold    | AHUNOLD | 590.423.4567      | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce      | Ernst     | BERNST | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David      | Austin    | DAUSTIN | 590.423.4569      | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli      | Pataballa | VPATABAL | 590.423.4560     | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana      | Lorentz   | DLORENTZ | 590.423.5567     | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy      | Greenberg | NGREENBE | 515.124.4569     | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel     | Faviet    | DFAVIET | 515.124.4169      | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John       | Chen      | JCHEN  | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael     | Sciarra   | ISCIARRA | 515.124.4369     | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman    | JMURMAN | 515.124.4469      | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis       | Popp      | LPOPP  | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den        | Raphaely  | DRAPHEAL | 515.127.4561     | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander  | Khoo      | AKHOO  | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli     | Baida     | SBAIDA | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal      | Tobias    | STOBIAS | 515.127.4564      | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy        | Himuro    | GHIMURO | 515.127.4565      | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566     | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew    | Weiss     | MWEISS | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam       | Fripp     | AFRIPP | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam      | Kaufling  | PKAUFLIN | 650.123.3234     | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta     | Vollman   | SVOLLMAN | 650.123.4234     | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin      | Mourgos   | KMOURGOS | 650.123.5234     | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia      | Nayer     | JNAYER | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224     | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James      | Landry    | JLANDRY | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven     | Markle    | SMARKLE | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura      | Bissot    | LBISSOT | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson  | MATKINSO | 650.124.6234     | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James      | Marlow    | JAMRLOW | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ         | Olson     | TJOLSON | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason      | Mallin    | JMALLIN | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael    | Rogers    | MROGERS | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki         | Gee       | KGEE   | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634     | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske     | Ladwig    | RLADWIG | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen    | Stiles    | SSTILES | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John       | Seo       | JSEO   | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua     | Patel     | JPATEL | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna     | Rajs      | TRAJS  | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis     | Davies    | CDAVIES | 650.121.2994      | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall    | Matos     | RMATOS | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter      | Vargas    | PVARGAS | 650.121.2004      | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John       | Russell   | JRUSSEL | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen      | Partners  | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto    | Errazuriz | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald     | Cambrault | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni      | Zlotkey   | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter      | Tucker    | PTUCKER | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
```

Latihan No. 9

Judul
Pertanyaan diskusi 1: jumlah pegawai 107, mengapa hasil join hanya 106 pegawai?

Syntax

```
SELECT COUNT(*) AS jumlah_pegawai FROM employees;

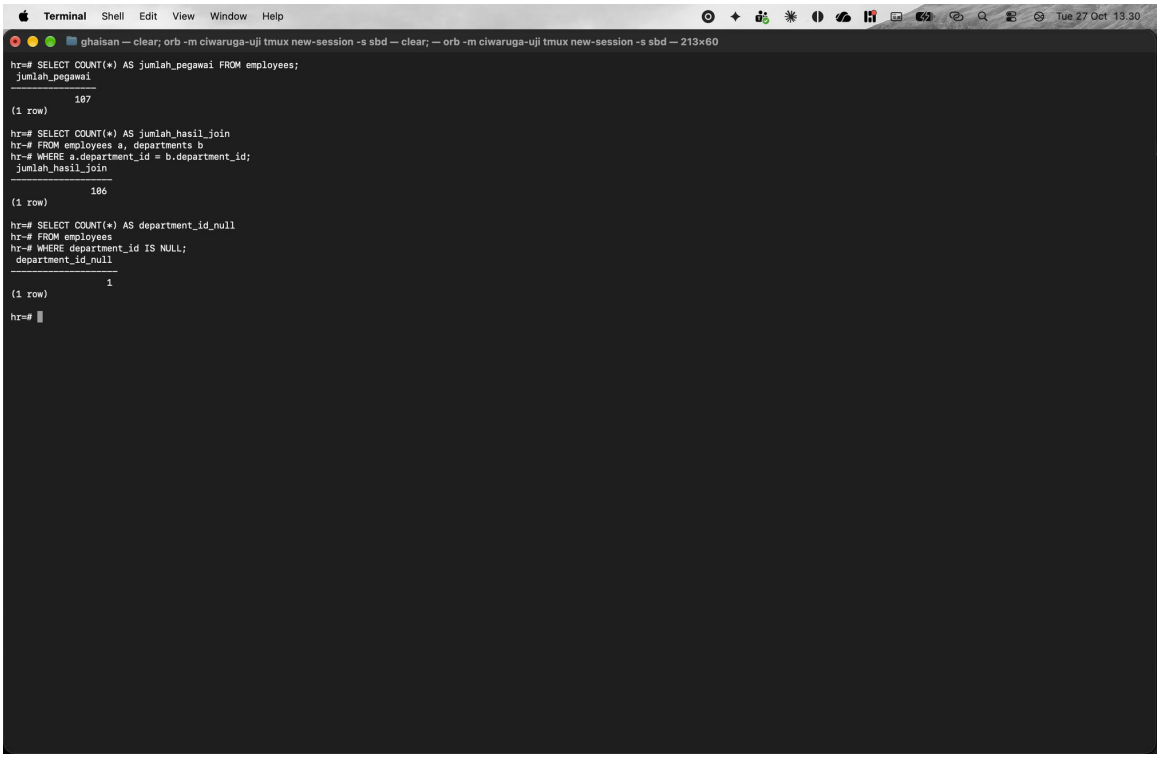
SELECT COUNT(*) AS jumlah_hasil_join
FROM employees a, departments b
WHERE a.department_id = b.department_id;

SELECT COUNT(*) AS department_id_null
FROM employees
WHERE department_id IS NULL;
```

Keterangan

Jawaban: tabel employees berisi 107 pegawai, tapi hasil join hanya 106 karena ada 1 pegawai yang department_id-nya NULL. Perbandingan NULL = angka tidak pernah bernilai benar, jadi baris itu tidak lolos kondisi join.

Hasil Eksekusi (Screenshot)



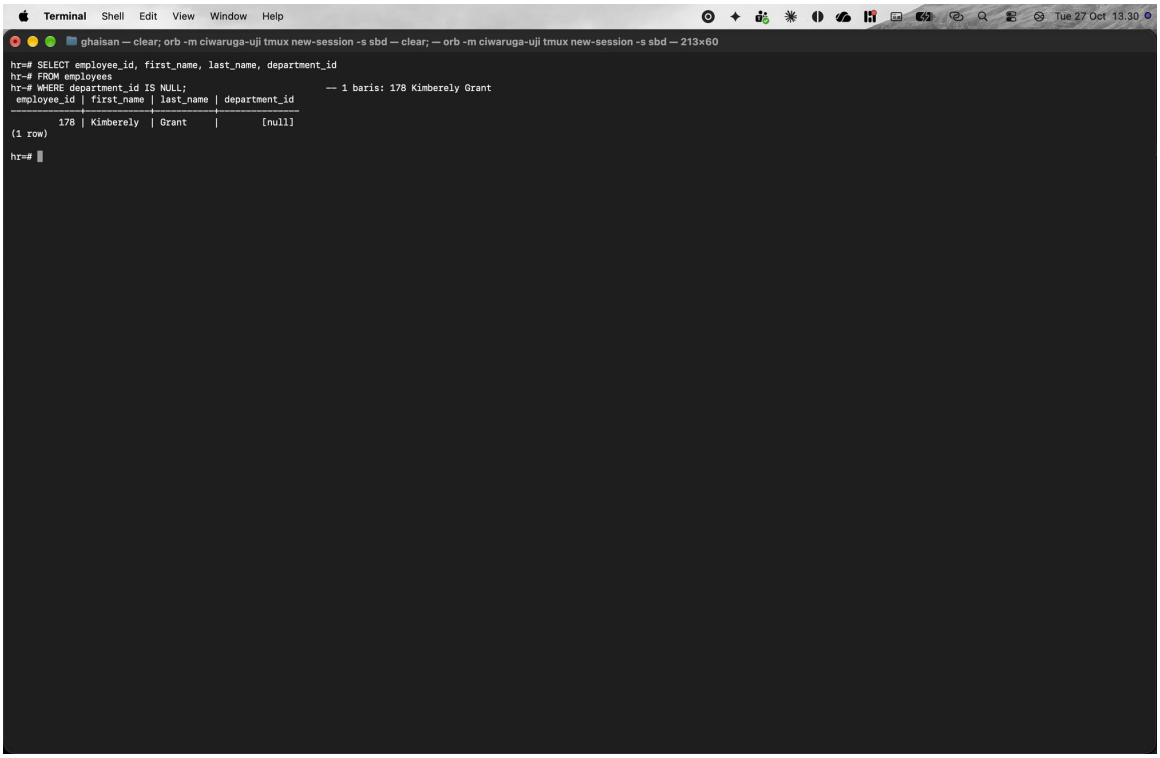
Latihan No. 10

Judul
Pertanyaan diskusi 2: siapakah pegawai yang belum ditempatkan di departemen?
Syntax

`SELECT employee_id, first_name, last_name, department_id`
`FROM employees`
`WHERE department_id IS NULL;  -- 1 baris: 178 Kimberely Grant`

Keterangan
Jawaban: pegawai nomor 178, Kimberely Grant, dengan department_id [null]. Untuk mencari NULL harus memakai IS NULL, bukan = NULL.

Hasil Eksekusi (Screenshot)

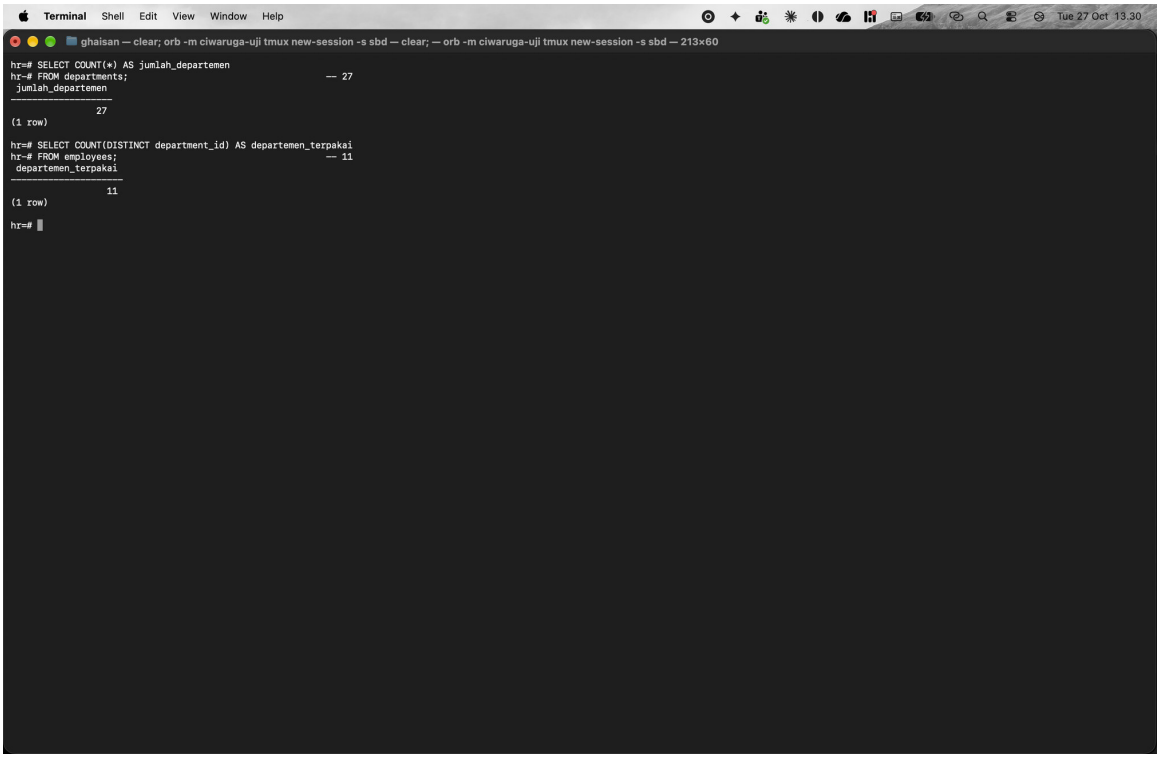


Latihan No. 11

Judul
Pertanyaan diskusi 3: apakah semua departemen sudah memiliki pegawai?
Syntax

`SELECT COUNT(*) AS jumlah_departemen`
`FROM departments;  -- 27`
 	
`SELECT COUNT(DISTINCT department_id) AS departemen_terpakai`
`FROM employees;  -- 11`

Keterangan
Jawaban: belum. Tabel departments berisi 27 departemen, tetapi pegawai hanya tersebar di 11 departemen (COUNT DISTINCT department_id di employees, NULL tidak ikut dihitung). Jadi ada 27 - 11 = 16 departemen yang belum punya pegawai. Departemen mana saja yang kosong dicari di bagian LEFT JOIN.

Hasil Eksekusi (Screenshot)


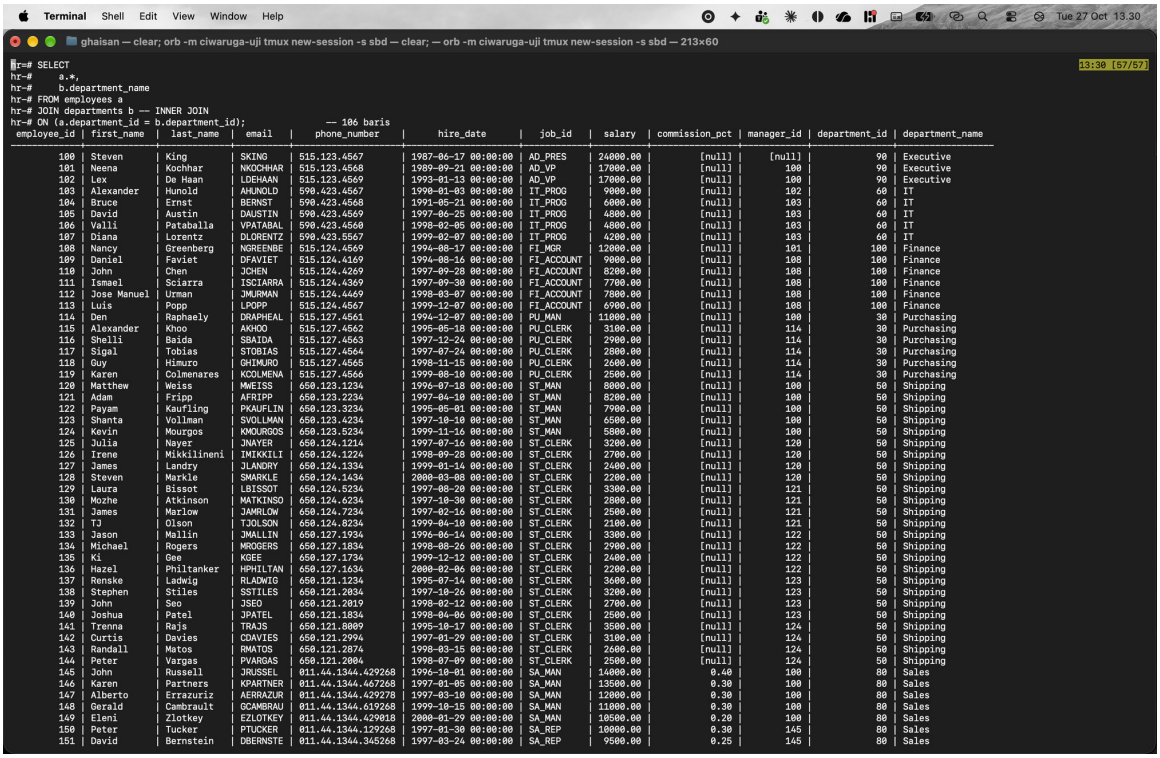
Latihan No. 12

Judul
[INNER] JOIN: employees JOIN departments
Syntax

`SELECT`
`  a.*,`
`  b.department_name`
`FROM employees a`
`JOIN departments b -- INNER JOIN`
`ON (a.department_id = b.department_id);  -- 106 baris`

Keterangan
JOIN sama dengan INNER JOIN, kata INNER boleh tidak ditulis. Kondisi join ditulis di ON, bukan di WHERE. Hasilnya 106 baris, sama dengan join memakai WHERE. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 27 Oct 13.30
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT                                                                                                    13:30 [57/57]
hr-#     a.*,
hr-#     b.department_name
hr-# FROM employees a
hr-# JOIN departments b -- INNER JOIN
hr-# ON (a.department_id = b.department_id);                -- 106 baris
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   |  salary  | commission_pct | manager_id | department_id | department_name
-------------+------------+------------+----------+--------------------+---------------------+------------+----------+----------------+------------+---------------+-----------------
         100 | Steven     | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 | [null]         | [null]     |            90 | Executive
         101 | Neena      | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 | [null]         |        100 |            90 | Executive
         102 | Lex        | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 | [null]         |        100 |            90 | Executive
         103 | Alexander  | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 | [null]         |        102 |            60 | IT
         104 | Bruce      | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 | [null]         |        103 |            60 | IT
         105 | David      | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 | [null]         |        103 |            60 | IT
         106 | Valli      | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 | [null]         |        103 |            60 | IT
         107 | Diana      | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 | [null]         |        103 |            60 | IT
         108 | Nancy      | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 | [null]         |        101 |           100 | Finance
         109 | Daniel     | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 | [null]         |        108 |           100 | Finance
         110 | John       | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 | [null]         |        108 |           100 | Finance
         111 | Ismael     | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 | [null]         |        108 |           100 | Finance
         112 | Jose Manuel | Urman     | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 | [null]         |        108 |           100 | Finance
         113 | Luis       | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 | [null]         |        108 |           100 | Finance
         114 | Den        | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 | [null]         |        100 |            30 | Purchasing
         115 | Alexander  | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 | [null]         |        114 |            30 | Purchasing
         116 | Shelli     | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 | [null]         |        114 |            30 | Purchasing
         117 | Sigal      | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 | [null]         |        114 |            30 | Purchasing
         118 | Guy        | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 | [null]         |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 | [null]         |        114 |            30 | Purchasing
         120 | Matthew    | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 | [null]         |        100 |            50 | Shipping
         121 | Adam       | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 | [null]         |        100 |            50 | Shipping
         122 | Payam      | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 | [null]         |        100 |            50 | Shipping
         123 | Shanta     | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 | [null]         |        100 |            50 | Shipping
         124 | Kevin      | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 | [null]         |        100 |            50 | Shipping
         125 | Julia      | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 | [null]         |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 | [null]         |        120 |            50 | Shipping
         127 | James      | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 | [null]         |        120 |            50 | Shipping
         128 | Steven     | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 | [null]         |        120 |            50 | Shipping
         129 | Laura      | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 | [null]         |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 | [null]         |        121 |            50 | Shipping
         131 | James      | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 | [null]         |        121 |            50 | Shipping
         132 | TJ         | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 | [null]         |        121 |            50 | Shipping
         133 | Jason      | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 | [null]         |        122 |            50 | Shipping
         134 | Michael    | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 | [null]         |        122 |            50 | Shipping
         135 | Ki         | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 | [null]         |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 | [null]         |        122 |            50 | Shipping
         137 | Renske     | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 | [null]         |        123 |            50 | Shipping
         138 | Stephen    | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 | [null]         |        123 |            50 | Shipping
         139 | John       | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 | [null]         |        123 |            50 | Shipping
         140 | Joshua     | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 | [null]         |        123 |            50 | Shipping
         141 | Trenna     | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 | [null]         |        124 |            50 | Shipping
         142 | Curtis     | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 | [null]         |        124 |            50 | Shipping
         143 | Randall    | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 | [null]         |        124 |            50 | Shipping
         144 | Peter      | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 | [null]         |        124 |            50 | Shipping
         145 | John       | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 | 0.40           |        100 |            80 | Sales
         146 | Karen      | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 | 0.30           |        100 |            80 | Sales
         147 | Alberto    | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 | 0.30           |        100 |            80 | Sales
         148 | Gerald     | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 | 0.30           |        100 |            80 | Sales
         149 | Eleni      | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 | 0.20           |        100 |            80 | Sales
         150 | Peter      | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 | 0.30           |        145 |            80 | Sales
         151 | David      | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 | 0.25           |        145 |            80 | Sales
```

Latihan No. 13

Judul
[INNER] JOIN: urutan tabel dibalik

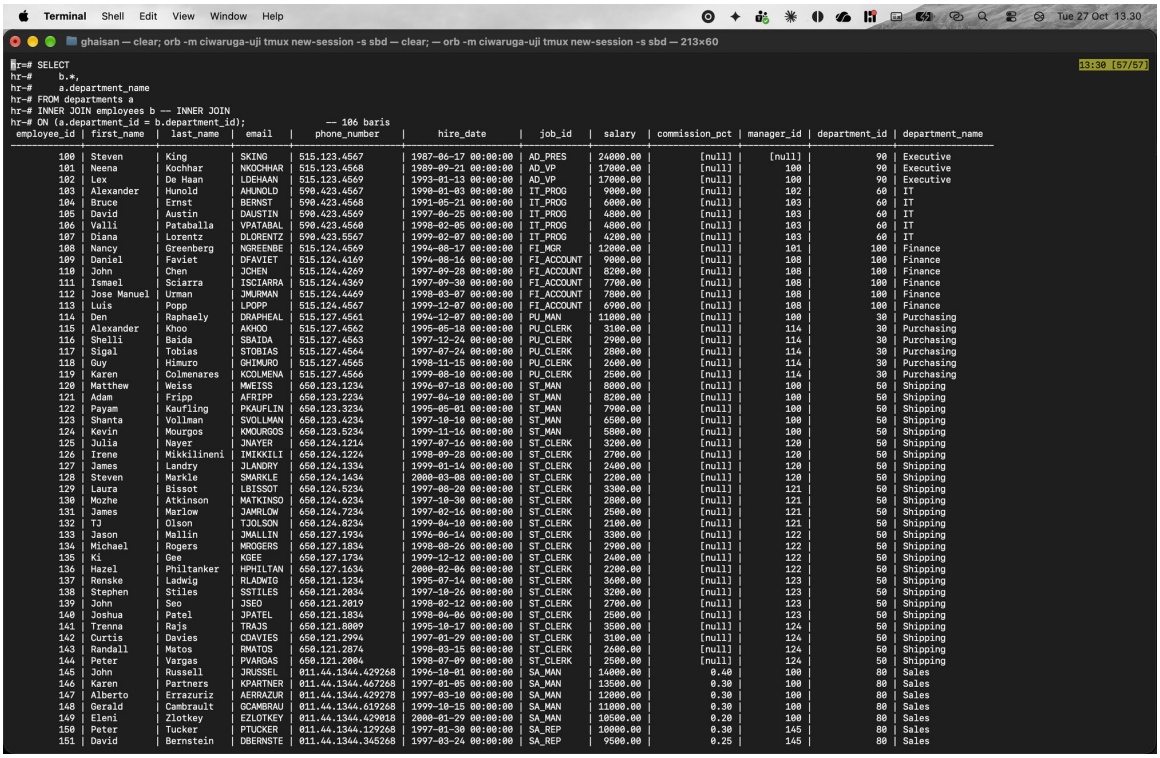
Syntax

```
SELECT
    b.*,
    a.department_name
FROM departments a
INNER JOIN employees b -- INNER JOIN
ON (a.department_id = b.department_id);                -- 106 baris
```

Keterangan

Urutan tabel dibalik (departments a INNER JOIN employees b), hasilnya tetap 106 baris.
Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.30
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT                                                                                              13:30 [57/57]
hr-#     b.*,
hr-#     a.department_name
hr-# FROM departments a
hr-# INNER JOIN employees b — INNER JOIN
hr-# ON (a.department_id = b.department_id);                    — 106 baris
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   | salary | commission_pct | manager_id | department_id | department_name
-------------+-------------+------------+----------+--------------------+---------------------+------------+--------+----------------+------------+---------------+-----------------
         100 | Steven      | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena       | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex         | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander   | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce       | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David       | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli       | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana       | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy       | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel      | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John        | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael      | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman      | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis        | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den         | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander   | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli      | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal       | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy         | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen       | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew     | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam        | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam       | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta      | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin       | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia       | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene       | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James       | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven      | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura       | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe       | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James       | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ          | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason       | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael     | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki          | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel       | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske      | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen     | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John        | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua      | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna      | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis      | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall     | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter       | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John        | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen       | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto     | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald      | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni       | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter       | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
         151 | David       | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 |           0.25 |        145 |            80 | Sales
```

Latihan No. 14

Judul
[INNER] JOIN: membandingkan waktu eksekusi join WHERE dan INNER JOIN

Syntax

```
EXPLAIN ANALYZE
SELECT a.*, b.department_name
FROM employees a, departments b
WHERE a.department_id = b.department_id;

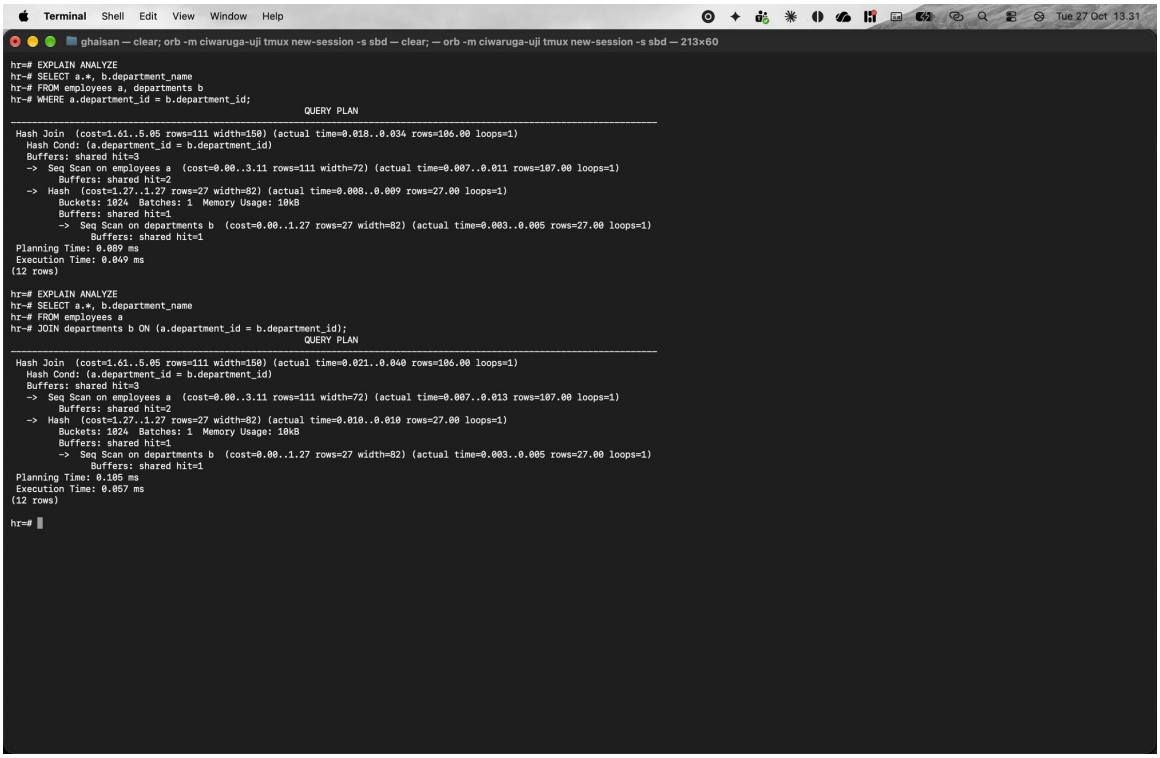
EXPLAIN ANALYZE
SELECT a.*, b.department_name
FROM employees a
JOIN departments b ON (a.department_id = b.department_id);
```

Keterangan

Pertanyaan slide: apakah join lebih cepat? EXPLAIN ANALYZE menampilkan rencana eksekusi dan waktu sebenarnya. Kedua query menghasilkan rencana yang sama persis (Hash Join, Seq Scan pada employees, Hash dari departments) dengan Execution Time yang sama-sama di bawah 0,1 ms.

Jadi PostgreSQL menjalankan join dengan WHERE dan INNER JOIN dengan cara yang sama. Selisih waktu di slide (0,180 dan 0,120 detik di pgAdmin) lebih dipengaruhi waktu menampilkan hasil, bukan cara join-nya. Kelebihan JOIN ... ON ada di keterbacaan, karena kondisi join terpisah dari kondisi penyaring di WHERE.

Hasil Eksekusi (Screenshot)



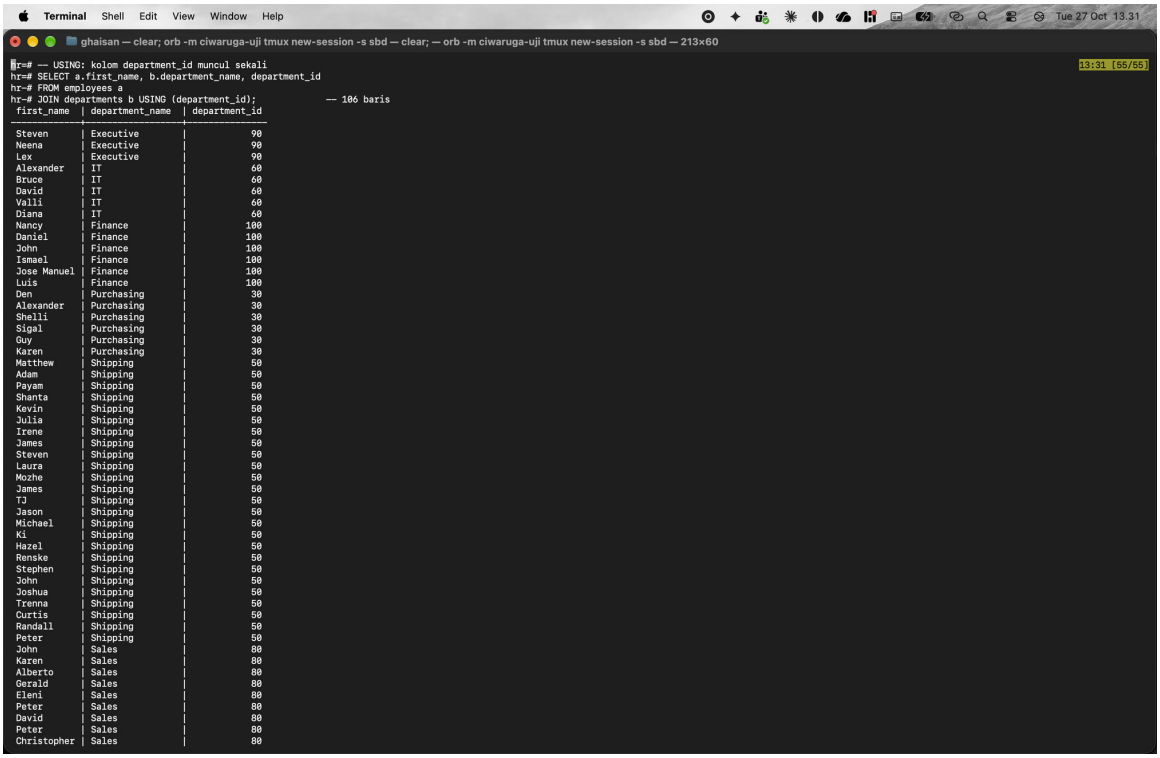
Latihan No. 15

Judul
Variasi USING: kolom department_id hanya muncul sekali
Syntax

`-- USING: kolom department_id muncul sekali`
`SELECT a.first_name, b.department_name, department_id`
`FROM employees a`
`JOIN departments b USING (department_id);  -- 106 baris`

Keterangan
USING (department_id) bisa dipakai kalau nama kolom penghubung sama di kedua tabel. Kolom department_id cukup ditulis tanpa alias dan hanya muncul sekali di hasil. Hasilnya 106 baris. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# -- USING: kolom department_id muncul sekali                          13:31 [55/55]
hr=# SELECT a.first_name, b.department_name, department_id
hr=# FROM employees a
hr-# JOIN departments b USING (department_id);            -- 106 baris
 first_name  | department_name  | department_id
-------------+------------------+--------------
 Steven      | Executive        |            90
 Neena       | Executive        |            90
 Lex         | Executive        |            90
 Alexander   | IT               |            60
 Bruce       | IT               |            60
 David       | IT               |            60
 Valli       | IT               |            60
 Diana       | IT               |            60
 Nancy       | Finance          |           100
 Daniel      | Finance          |           100
 John        | Finance          |           100
 Ismael      | Finance          |           100
 Jose Manuel | Finance          |           100
 Luis        | Finance          |           100
 Den         | Purchasing       |            30
 Alexander   | Purchasing       |            30
 Shelli      | Purchasing       |            30
 Sigal       | Purchasing       |            30
 Guy         | Purchasing       |            30
 Karen       | Purchasing       |            30
 Matthew     | Shipping         |            50
 Adam        | Shipping         |            50
 Payam       | Shipping         |            50
 Shanta      | Shipping         |            50
 Kevin       | Shipping         |            50
 Julia       | Shipping         |            50
 Irene       | Shipping         |            50
 James       | Shipping         |            50
 Steven      | Shipping         |            50
 Laura       | Shipping         |            50
 Mozhe       | Shipping         |            50
 James       | Shipping         |            50
 TJ          | Shipping         |            50
 Jason       | Shipping         |            50
 Michael     | Shipping         |            50
 Ki          | Shipping         |            50
 Hazel       | Shipping         |            50
 Renske      | Shipping         |            50
 Stephen     | Shipping         |            50
 John        | Shipping         |            50
 Joshua      | Shipping         |            50
 Trenna      | Shipping         |            50
 Curtis      | Shipping         |            50
 Randall     | Shipping         |            50
 Peter       | Shipping         |            50
 John        | Sales            |            80
 Karen       | Sales            |            80
 Alberto     | Sales            |            80
 Gerald      | Sales            |            80
 Eleni       | Sales            |            80
 Peter       | Sales            |            80
 David       | Sales            |            80
 Peter       | Sales            |            80
 Christopher | Sales            |            80
```

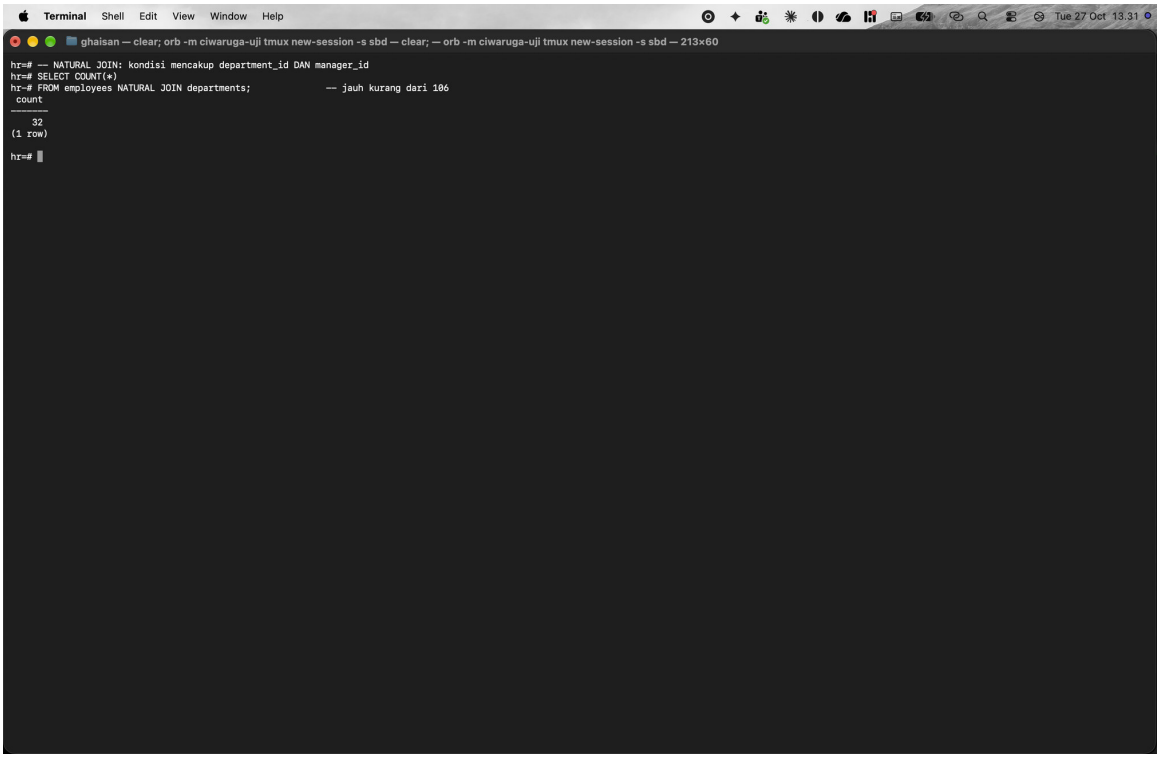
Latihan No. 16

Judul
Variasi NATURAL JOIN: kondisi join mencakup department_id dan manager_id
Syntax

`-- NATURAL JOIN: kondisi mencakup department_id DAN manager_id`
`SELECT COUNT(*)`
`FROM employees NATURAL JOIN departments;  -- jauh kurang dari 106`

Keterangan
NATURAL JOIN otomatis menggabungkan semua kolom yang namanya sama. employees dan departments sama-sama punya department_id dan manager_id, jadi kondisinya menjadi department_id sama DAN manager_id sama. Hasilnya hanya 32 baris, bukan 106, karena yang lolos cuma pegawai yang atasannya kebetulan manajer departemennya sendiri. Karena itu join tetap ditulis dengan ON.

Hasil Eksekusi (Screenshot)



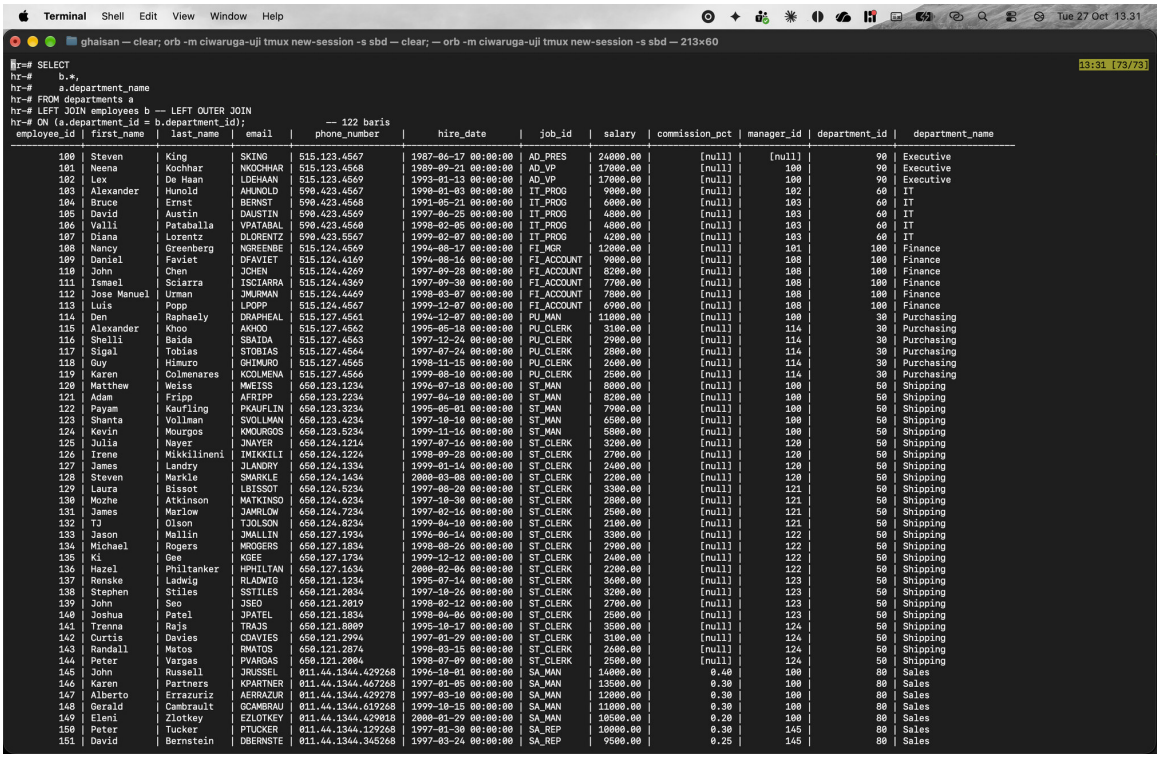
Latihan No. 17

Judul
LEFT [OUTER] JOIN: semua departemen beserta pegawainya
Syntax

`SELECT`
`  b.*,`
`  a.department_name`
`FROM departments a`
`LEFT JOIN employees b -- LEFT OUTER JOIN`
`ON (a.department_id = b.department_id);  -- 122 baris`

Keterangan
LEFT JOIN mengambil semua baris tabel kiri (departments). Hasilnya 122 baris: 106 pasangan departemen dengan pegawai ditambah 16 departemen tanpa pegawai. Pada 16 baris terakhir semua kolom employees bernilai [null], baris-baris ini ditampilkan khusus di latihan 19. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT                                                                                              13:31 [73/73]
hr-#     b.*,
hr-#     a.department_name
hr-# FROM departments a
hr-# LEFT JOIN employees b -- LEFT OUTER JOIN
hr-# ON (a.department_id = b.department_id);                -- 122 baris
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   | salary | commission_pct | manager_id | department_id |   department_name
-------------+------------+------------+----------+--------------------+---------------------+------------+--------+----------------+------------+---------------+---------------------
         100 | Steven     | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena      | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex        | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander  | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce      | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David      | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli      | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana      | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy      | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel     | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John       | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael     | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman     | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis       | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den        | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander  | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli     | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal      | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy        | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew    | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam       | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam      | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta     | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin      | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia      | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James      | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven     | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura      | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James      | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ         | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason      | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael    | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki         | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske     | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen    | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John       | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua     | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna     | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis     | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall    | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter      | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John       | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen      | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto    | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald     | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni      | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter      | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
         151 | David      | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 |           0.25 |        145 |            80 | Sales
```

Latihan No. 18

Judul
LEFT [OUTER] JOIN: syntax outer join gaya lama Oracle (+) di PostgreSQL

Syntax

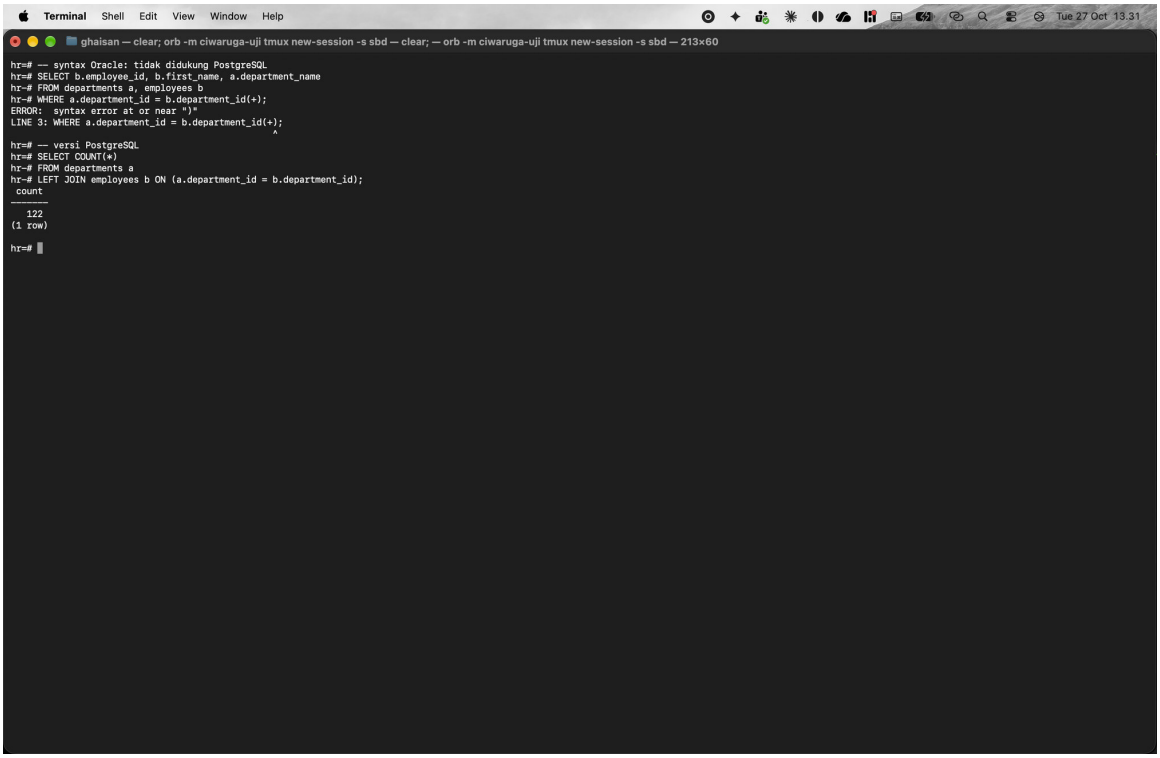
```
-- syntax Oracle: tidak didukung PostgreSQL
SELECT b.employee_id, b.first_name, a.department_name
FROM departments a, employees b
WHERE a.department_id = b.department_id(+);

-- versi PostgreSQL
SELECT COUNT(*)
FROM departments a
LEFT JOIN employees b ON (a.department_id = b.department_id);
```

Keterangan

Materi Oracle lama memakai outer join gaya lama dengan tanda (+) di WHERE. PostgreSQL tidak mengenal syntax ini sehingga muncul syntax error. Penggantinya LEFT JOIN ... ON, hasilnya 122 baris seperti latihan sebelumnya.

Hasil Eksekusi (Screenshot)



Latihan No. 19

Judul
LEFT [OUTER] JOIN: departemen yang belum punya pegawai

Syntax

```
SELECT
    b.*,
    a.department_name
FROM departments a
LEFT JOIN employees b -- LEFT OUTER JOIN
ON (a.department_id = b.department_id)
WHERE b.employee_id IS NULL;                          -- 16 baris

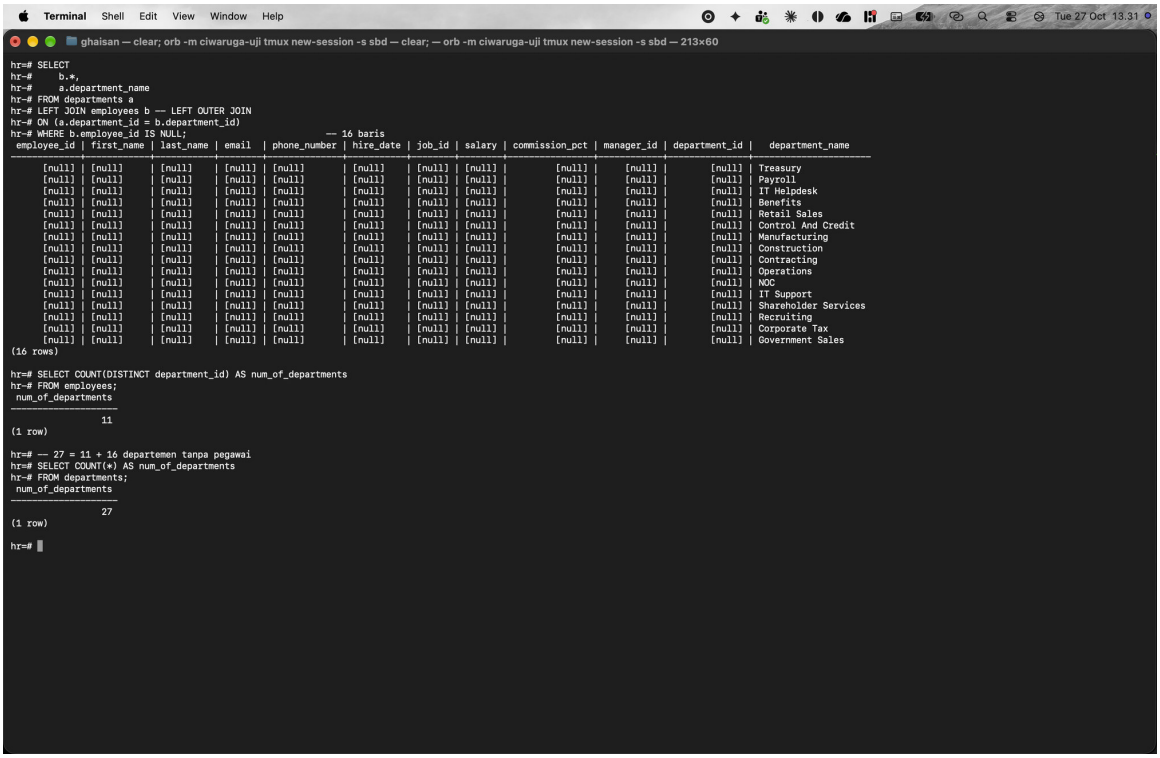
SELECT COUNT(DISTINCT department_id) AS num_of_departments
FROM employees;
-- 27 = 11 + 16 departemen tanpa pegawai

SELECT COUNT(*) AS num_of_departments
FROM departments;
```

Keterangan

WHERE b.employee_id IS NULL menyisakan baris yang tidak punya pasangan, karena employee_id adalah primary key yang tidak pernah NULL kalau barisnya punya pasangan. Hasilnya 16 departemen, di antaranya Treasury, Payroll, dan Benefits.
Cocok dengan pembuktian jumlah: 27 departemen = 11 departemen yang punya pegawai + 16 departemen tanpa pegawai.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT
hr-#     b.*,
hr-#     a.department_name
hr-# FROM departments a
hr-# LEFT JOIN employees b -- LEFT OUTER JOIN
hr-# ON (a.department_id = b.department_id)
hr-# WHERE b.employee_id IS NULL;                                  -- 16 baris
 employee_id | first_name | last_name | email  | phone_number | hire_date | job_id | salary | commission_pct | manager_id | department_id |   department_name
-------------+------------+-----------+--------+--------------+-----------+--------+--------+----------------+------------+---------------+---------------------
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Treasury
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Payroll
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | IT Helpdesk
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Benefits
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Retail Sales
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Control And Credit
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Manufacturing
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Construction
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Contracting
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Operations
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | NOC
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | IT Support
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Shareholder Services
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Recruiting
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Corporate Tax
      [null] | [null]     | [null]    | [null] | [null]       | [null]    | [null] | [null] |         [null] |     [null] |        [null] | Government Sales
(16 rows)

hr=# SELECT COUNT(DISTINCT department_id) AS num_of_departments
hr-# FROM employees;
 num_of_departments
--------------------
                 11
(1 row)

hr=# -- 27 = 11 + 16 departemen tanpa pegawai
hr=# SELECT COUNT(*) AS num_of_departments
hr-# FROM departments;
 num_of_departments
--------------------
                 27
(1 row)

hr=#
```

Latihan No. 20

Judul
LEFT [OUTER] JOIN: syarat biasa pada kolom tabel kanan membuat LEFT JOIN seperti INNER JOIN

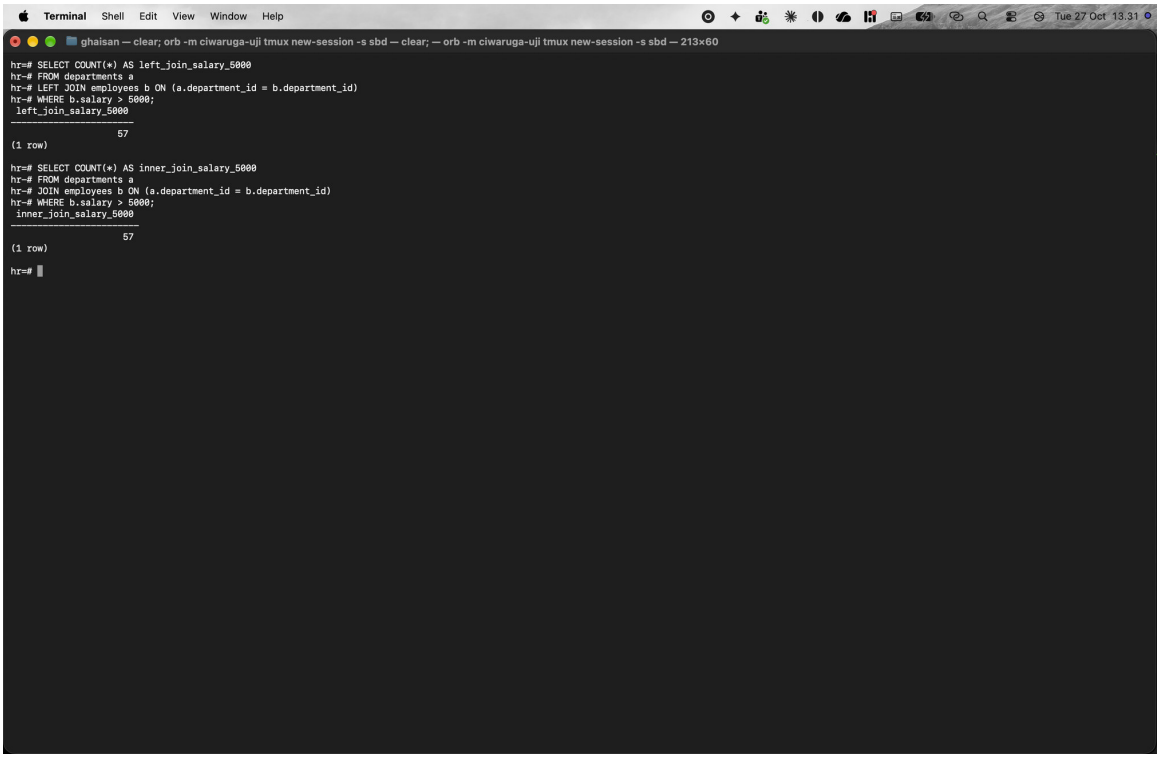
Syntax

```
SELECT COUNT(*) AS left_join_salary_5000
FROM departments a
LEFT JOIN employees b ON (a.department_id = b.department_id)
WHERE b.salary > 5000;

SELECT COUNT(*) AS inner_join_salary_5000
FROM departments a
JOIN employees b ON (a.department_id = b.department_id)
WHERE b.salary > 5000;
```

Keterangan

Percobaan tambahan dari catatan slide. Kalau kolom tabel kanan disaring dengan syarat biasa (b.salary > 5000), baris departemen tanpa pegawai ikut terbuang karena NULL > 5000 tidak bernilai benar. Hasil LEFT JOIN jadi 57 baris, sama persis dengan INNER JOIN. Satu-satunya syarat di WHERE yang tetap mempertahankan baris tanpa pasangan adalah IS NULL.

Hasil Eksekusi (Screenshot)


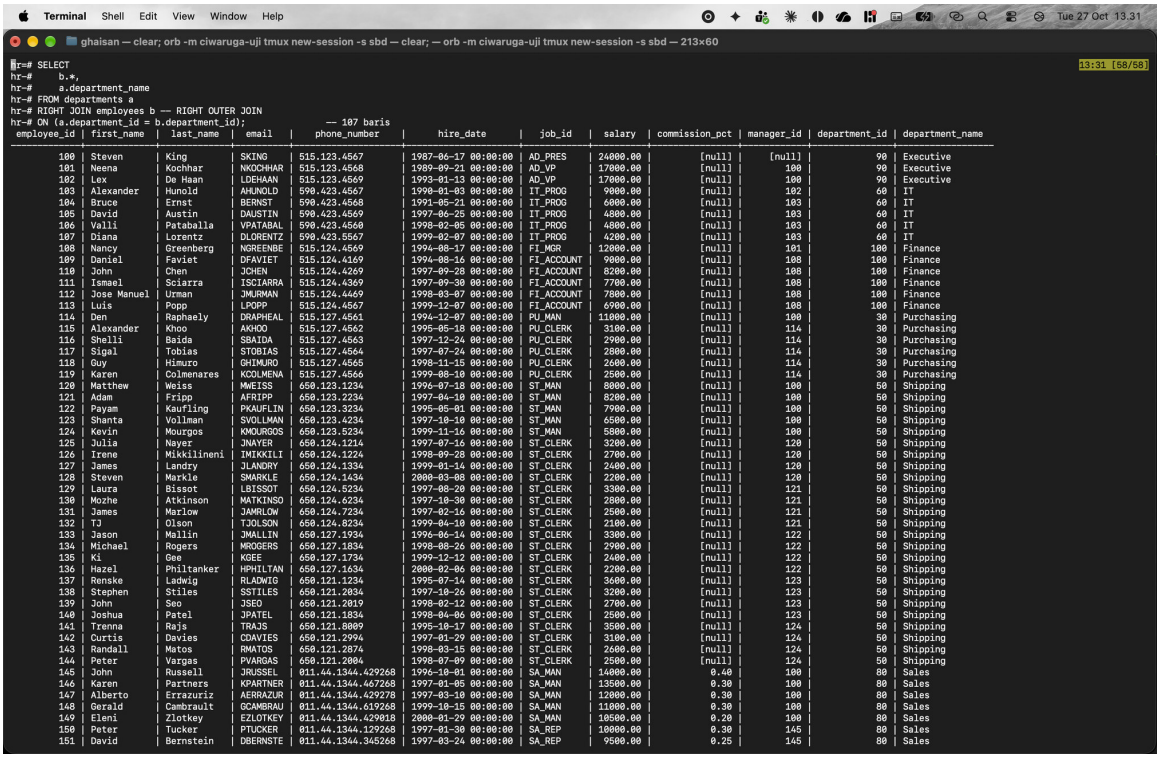
Latihan No. 21

Judul
RIGHT [OUTER] JOIN: semua pegawai beserta departemennya
Syntax

`SELECT`
`  b.*,`
`  a.department_name`
`FROM departments a`
`RIGHT JOIN employees b -- RIGHT OUTER JOIN`
`ON (a.department_id = b.department_id);  -- 107 baris`

Keterangan
RIGHT JOIN mengambil semua baris tabel kanan (employees), jadi hasilnya 107 baris. Semua pegawai tampil, termasuk Kimberely Grant yang department_id dan department_name-nya [null]. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60

hr=# SELECT                                                                                              13:31 [58/58]
hr-#     b.*,
hr-#     a.department_name
hr-# FROM departments a
hr-# RIGHT JOIN employees b -- RIGHT OUTER JOIN
hr-# ON (a.department_id = b.department_id);                  -- 107 baris
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   | salary  | commission_pct | manager_id | department_id | department_name
-------------+------------+------------+----------+--------------------+---------------------+------------+---------+----------------+------------+---------------+-----------------
         100 | Steven     | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena      | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex        | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander  | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce      | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David      | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli      | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana      | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy      | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel     | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John       | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael     | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman     | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis       | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den        | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander  | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli     | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal      | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy        | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew    | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam       | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam      | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta     | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin      | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia      | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James      | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven     | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura      | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James      | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ         | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason      | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael    | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki         | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske     | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen    | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John       | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua     | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna     | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis     | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall    | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter      | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John       | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen      | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto    | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald     | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni      | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter      | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
         151 | David      | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 |           0.25 |        145 |            80 | Sales
```

Latihan No. 22

Judul
RIGHT [OUTER] JOIN: pegawai yang belum ditempatkan di departemen

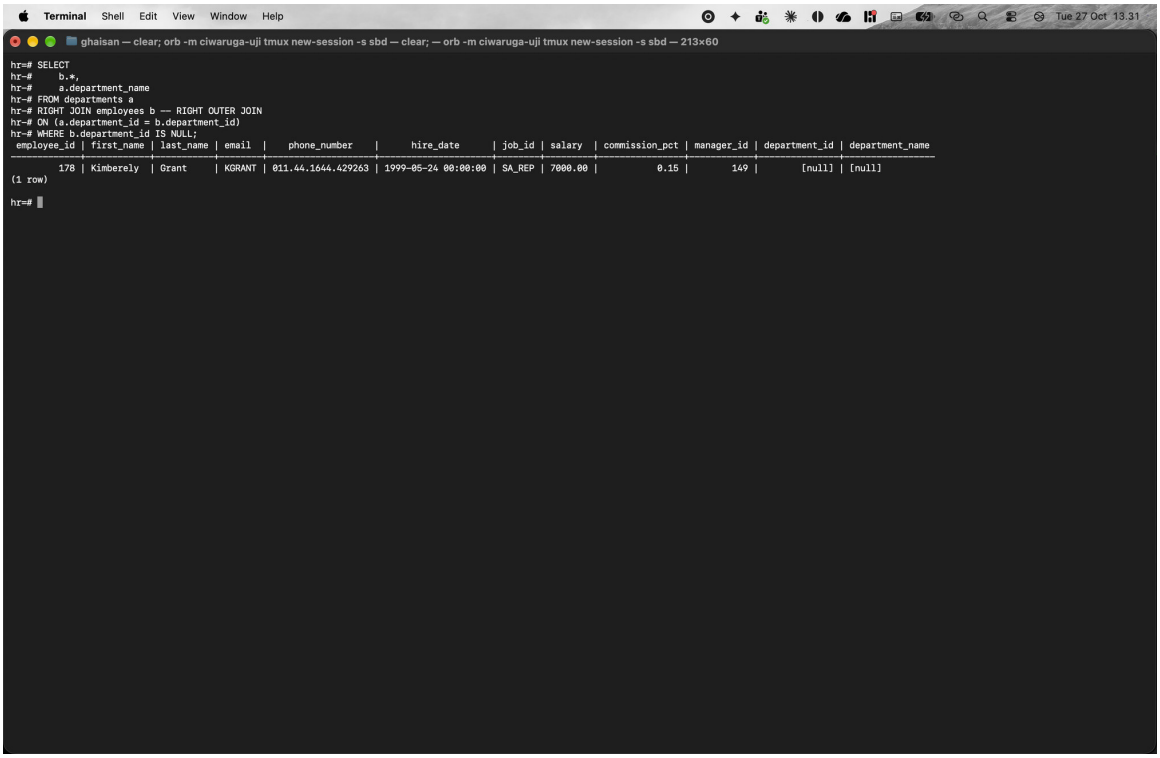
Syntax

```
SELECT
    b.*,
    a.department_name
FROM departments a
RIGHT JOIN employees b -- RIGHT OUTER JOIN
ON (a.department_id = b.department_id)
WHERE b.department_id IS NULL;
```

Keterangan

Dengan WHERE b.department_id IS NULL tersisa 1 baris, yaitu pegawai 178 Kimberely Grant.

Hasil Eksekusi (Screenshot)



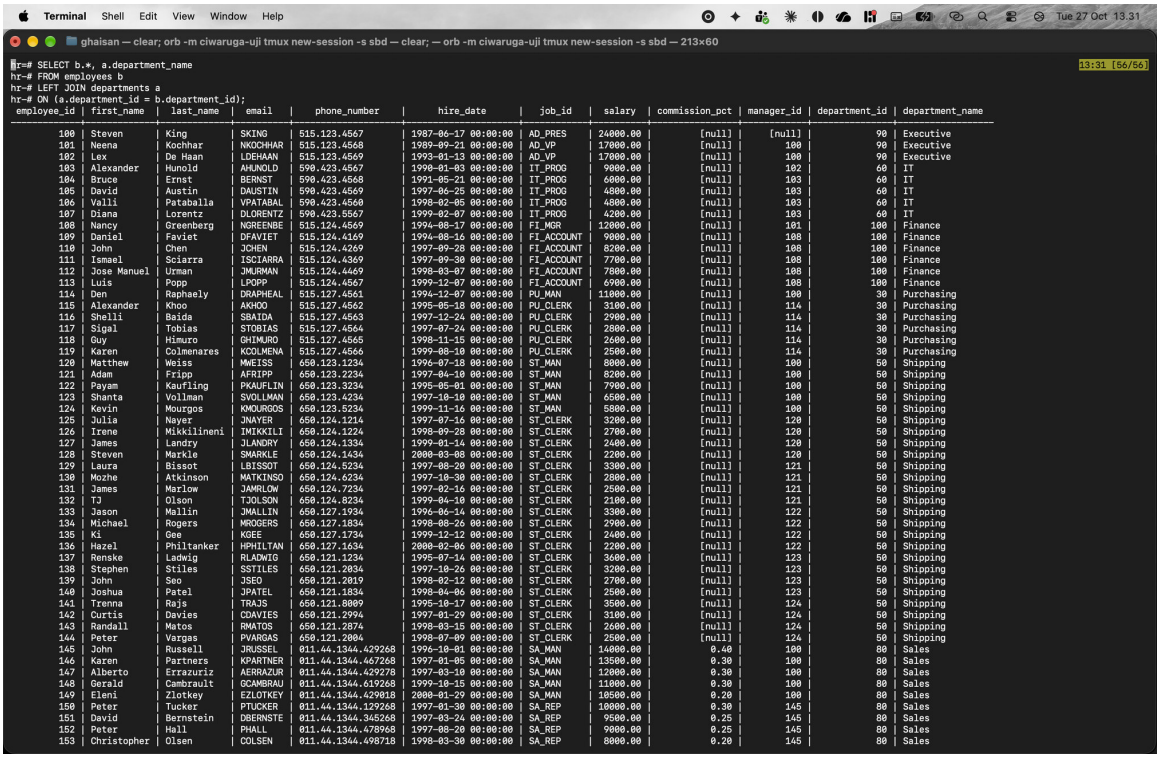
Latihan No. 23

Judul
RIGHT [OUTER] JOIN ditulis ulang sebagai LEFT JOIN dengan urutan tabel dibalik
Syntax

`SELECT b.*, a.department_name`
`FROM employees b`
`LEFT JOIN departments a`
`ON (a.department_id = b.department_id);`

Keterangan
RIGHT JOIN selalu bisa ditulis ulang jadi LEFT JOIN dengan urutan tabel dibalik. employees LEFT JOIN departments menghasilkan 107 baris, sama dengan RIGHT JOIN sebelumnya. Dalam praktik LEFT JOIN lebih sering dipakai karena tabel utamanya ditulis paling kiri. Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# SELECT b.*, a.department_name                                                    13:31 [56/56]
hr-# FROM employees b
hr-# LEFT JOIN departments a
hr-# ON (a.department_id = b.department_id);
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   | salary  | commission_pct | manager_id | department_id | department_name
-------------+-------------+------------+----------+--------------------+---------------------+------------+---------+----------------+------------+---------------+-----------------
         100 | Steven      | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 |         [null] |     [null] |            90 | Executive
         101 | Neena       | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         102 | Lex         | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 |         [null] |        100 |            90 | Executive
         103 | Alexander   | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 |         [null] |        102 |            60 | IT
         104 | Bruce       | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 |         [null] |        103 |            60 | IT
         105 | David       | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         106 | Valli       | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 |         [null] |        103 |            60 | IT
         107 | Diana       | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 |         [null] |        103 |            60 | IT
         108 | Nancy       | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 |         [null] |        101 |           100 | Finance
         109 | Daniel      | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 |         [null] |        108 |           100 | Finance
         110 | John        | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 |         [null] |        108 |           100 | Finance
         111 | Ismael      | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 |         [null] |        108 |           100 | Finance
         112 | Jose Manuel | Urman      | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 |         [null] |        108 |           100 | Finance
         113 | Luis        | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 |         [null] |        108 |           100 | Finance
         114 | Den         | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 |         [null] |        100 |            30 | Purchasing
         115 | Alexander   | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 |         [null] |        114 |            30 | Purchasing
         116 | Shelli      | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 |         [null] |        114 |            30 | Purchasing
         117 | Sigal       | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 |         [null] |        114 |            30 | Purchasing
         118 | Guy         | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 |         [null] |        114 |            30 | Purchasing
         119 | Karen       | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 |         [null] |        114 |            30 | Purchasing
         120 | Matthew     | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 |         [null] |        100 |            50 | Shipping
         121 | Adam        | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 |         [null] |        100 |            50 | Shipping
         122 | Payam       | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 |         [null] |        100 |            50 | Shipping
         123 | Shanta      | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 |         [null] |        100 |            50 | Shipping
         124 | Kevin       | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 |         [null] |        100 |            50 | Shipping
         125 | Julia       | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        120 |            50 | Shipping
         126 | Irene       | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        120 |            50 | Shipping
         127 | James       | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        120 |            50 | Shipping
         128 | Steven      | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        120 |            50 | Shipping
         129 | Laura       | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        121 |            50 | Shipping
         130 | Mozhe       | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 |         [null] |        121 |            50 | Shipping
         131 | James       | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        121 |            50 | Shipping
         132 | TJ          | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 |         [null] |        121 |            50 | Shipping
         133 | Jason       | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 |         [null] |        122 |            50 | Shipping
         134 | Michael     | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 |         [null] |        122 |            50 | Shipping
         135 | Ki          | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 |         [null] |        122 |            50 | Shipping
         136 | Hazel       | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 |         [null] |        122 |            50 | Shipping
         137 | Renske      | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 |         [null] |        123 |            50 | Shipping
         138 | Stephen     | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 |         [null] |        123 |            50 | Shipping
         139 | John        | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 |         [null] |        123 |            50 | Shipping
         140 | Joshua      | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        123 |            50 | Shipping
         141 | Trenna      | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 |         [null] |        124 |            50 | Shipping
         142 | Curtis      | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 |         [null] |        124 |            50 | Shipping
         143 | Randall     | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 |         [null] |        124 |            50 | Shipping
         144 | Peter       | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 |         [null] |        124 |            50 | Shipping
         145 | John        | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |           0.40 |        100 |            80 | Sales
         146 | Karen       | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |           0.30 |        100 |            80 | Sales
         147 | Alberto     | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |           0.30 |        100 |            80 | Sales
         148 | Gerald      | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |           0.30 |        100 |            80 | Sales
         149 | Eleni       | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |           0.20 |        100 |            80 | Sales
         150 | Peter       | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |           0.30 |        145 |            80 | Sales
         151 | David       | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 |           0.25 |        145 |            80 | Sales
         152 | Peter       | Hall       | PHALL    | 011.44.1344.478968 | 1997-08-20 00:00:00 | SA_REP     |  9000.00 |           0.25 |        145 |            80 | Sales
         153 | Christopher | Olsen      | COLSEN   | 011.44.1344.498718 | 1998-03-30 00:00:00 | SA_REP     |  8000.00 |           0.20 |        145 |            80 | Sales
```

Latihan No. 24

Judul
FULL [OUTER] JOIN: semua departemen dan semua pegawai

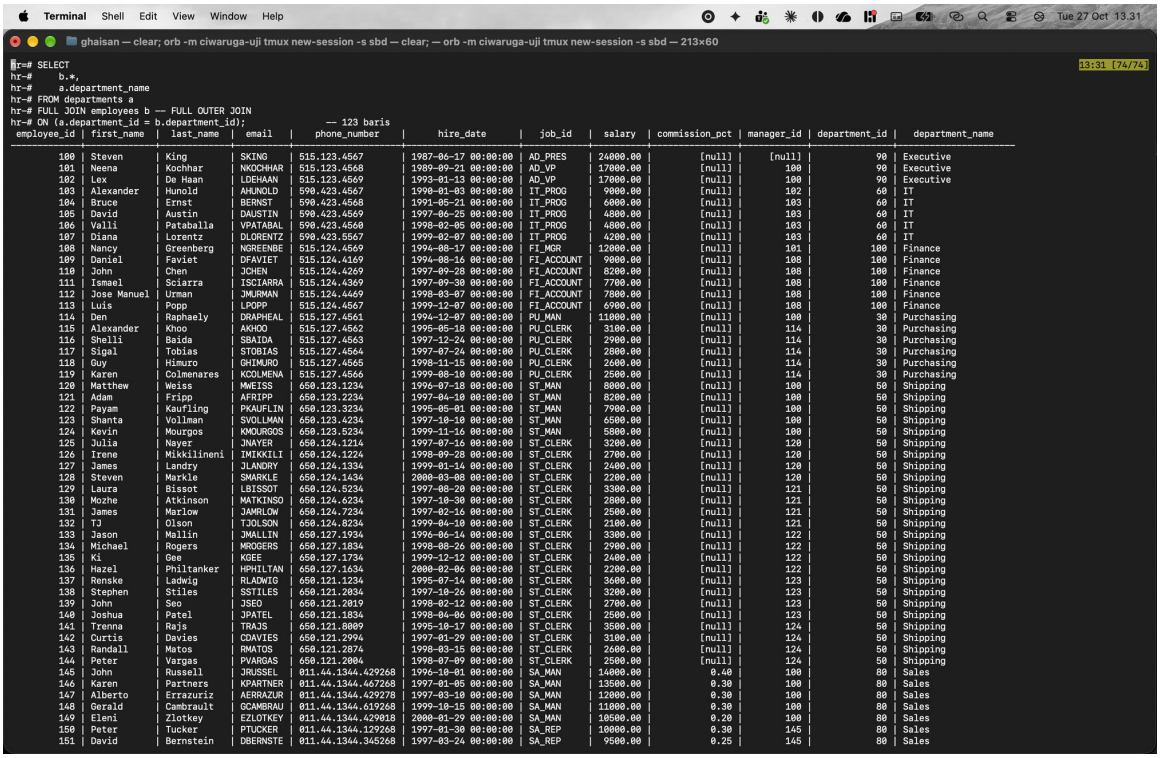
Syntax

```
SELECT
    b.*,
    a.department_name
FROM departments a
FULL JOIN employees b -- FULL OUTER JOIN
ON (a.department_id = b.department_id);                -- 123 baris
```

Keterangan

FULL JOIN mengambil semua baris dari kedua tabel. Hasilnya 123 baris = 106 pasangan + 16 departemen tanpa pegawai + 1 pegawai tanpa departemen.
Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                                    Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# SELECT                                                                                                  13:31 [74/74]
hr-#     b.*,
hr-#     a.department_name
hr-# FROM departments a
hr-# FULL JOIN employees b -- FULL OUTER JOIN
hr-# ON (a.department_id = b.department_id);                    -- 123 baris
 employee_id | first_name  | last_name  |  email   |    phone_number    |      hire_date      |   job_id   | salary | commission_pct | manager_id | department_id |   department_name
-------------+------------+------------+----------+--------------------+---------------------+------------+--------+----------------+------------+---------------+---------------------
         100 | Steven     | King       | SKING    | 515.123.4567       | 1987-06-17 00:00:00 | AD_PRES    | 24000.00 | [null]       | [null]     |            90 | Executive
         101 | Neena      | Kochhar    | NKOCHHAR | 515.123.4568       | 1989-09-21 00:00:00 | AD_VP      | 17000.00 | [null]       |        100 |            90 | Executive
         102 | Lex        | De Haan    | LDEHAAN  | 515.123.4569       | 1993-01-13 00:00:00 | AD_VP      | 17000.00 | [null]       |        100 |            90 | Executive
         103 | Alexander  | Hunold     | AHUNOLD  | 590.423.4567       | 1990-01-03 00:00:00 | IT_PROG    |  9000.00 | [null]       |        102 |            60 | IT
         104 | Bruce      | Ernst      | BERNST   | 590.423.4568       | 1991-05-21 00:00:00 | IT_PROG    |  6000.00 | [null]       |        103 |            60 | IT
         105 | David      | Austin     | DAUSTIN  | 590.423.4569       | 1997-06-25 00:00:00 | IT_PROG    |  4800.00 | [null]       |        103 |            60 | IT
         106 | Valli      | Pataballa  | VPATABAL | 590.423.4560       | 1998-02-05 00:00:00 | IT_PROG    |  4800.00 | [null]       |        103 |            60 | IT
         107 | Diana      | Lorentz    | DLORENTZ | 590.423.5567       | 1999-02-07 00:00:00 | IT_PROG    |  4200.00 | [null]       |        103 |            60 | IT
         108 | Nancy      | Greenberg  | NGREENBE | 515.124.4569       | 1994-08-17 00:00:00 | FI_MGR     | 12000.00 | [null]       |        101 |           100 | Finance
         109 | Daniel     | Faviet     | DFAVIET  | 515.124.4169       | 1994-08-16 00:00:00 | FI_ACCOUNT |  9000.00 | [null]       |        108 |           100 | Finance
         110 | John       | Chen       | JCHEN    | 515.124.4269       | 1997-09-28 00:00:00 | FI_ACCOUNT |  8200.00 | [null]       |        108 |           100 | Finance
         111 | Ismael     | Sciarra    | ISCIARRA | 515.124.4369       | 1997-09-30 00:00:00 | FI_ACCOUNT |  7700.00 | [null]       |        108 |           100 | Finance
         112 | Jose Manuel | Urman     | JMURMAN  | 515.124.4469       | 1998-03-07 00:00:00 | FI_ACCOUNT |  7800.00 | [null]       |        108 |           100 | Finance
         113 | Luis       | Popp       | LPOPP    | 515.124.4567       | 1999-12-07 00:00:00 | FI_ACCOUNT |  6900.00 | [null]       |        108 |           100 | Finance
         114 | Den        | Raphaely   | DRAPHEAL | 515.127.4561       | 1994-12-07 00:00:00 | PU_MAN     | 11000.00 | [null]       |        100 |            30 | Purchasing
         115 | Alexander  | Khoo       | AKHOO    | 515.127.4562       | 1995-05-18 00:00:00 | PU_CLERK   |  3100.00 | [null]       |        114 |            30 | Purchasing
         116 | Shelli     | Baida      | SBAIDA   | 515.127.4563       | 1997-12-24 00:00:00 | PU_CLERK   |  2900.00 | [null]       |        114 |            30 | Purchasing
         117 | Sigal      | Tobias     | STOBIAS  | 515.127.4564       | 1997-07-24 00:00:00 | PU_CLERK   |  2800.00 | [null]       |        114 |            30 | Purchasing
         118 | Guy        | Himuro     | GHIMURO  | 515.127.4565       | 1998-11-15 00:00:00 | PU_CLERK   |  2600.00 | [null]       |        114 |            30 | Purchasing
         119 | Karen      | Colmenares | KCOLMENA | 515.127.4566       | 1999-08-10 00:00:00 | PU_CLERK   |  2500.00 | [null]       |        114 |            30 | Purchasing
         120 | Matthew    | Weiss      | MWEISS   | 650.123.1234       | 1996-07-18 00:00:00 | ST_MAN     |  8000.00 | [null]       |        100 |            50 | Shipping
         121 | Adam       | Fripp      | AFRIPP   | 650.123.2234       | 1997-04-10 00:00:00 | ST_MAN     |  8200.00 | [null]       |        100 |            50 | Shipping
         122 | Payam      | Kaufling   | PKAUFLIN | 650.123.3234       | 1995-05-01 00:00:00 | ST_MAN     |  7900.00 | [null]       |        100 |            50 | Shipping
         123 | Shanta     | Vollman    | SVOLLMAN | 650.123.4234       | 1997-10-10 00:00:00 | ST_MAN     |  6500.00 | [null]       |        100 |            50 | Shipping
         124 | Kevin      | Mourgos    | KMOURGOS | 650.123.5234       | 1999-11-16 00:00:00 | ST_MAN     |  5800.00 | [null]       |        100 |            50 | Shipping
         125 | Julia      | Nayer      | JNAYER   | 650.124.1214       | 1997-07-16 00:00:00 | ST_CLERK   |  3200.00 | [null]       |        120 |            50 | Shipping
         126 | Irene      | Mikkilineni | IMIKKILI | 650.124.1224       | 1998-09-28 00:00:00 | ST_CLERK   |  2700.00 | [null]       |        120 |            50 | Shipping
         127 | James      | Landry     | JLANDRY  | 650.124.1334       | 1999-01-14 00:00:00 | ST_CLERK   |  2400.00 | [null]       |        120 |            50 | Shipping
         128 | Steven     | Markle     | SMARKLE  | 650.124.1434       | 2000-03-08 00:00:00 | ST_CLERK   |  2200.00 | [null]       |        120 |            50 | Shipping
         129 | Laura      | Bissot     | LBISSOT  | 650.124.5234       | 1997-08-20 00:00:00 | ST_CLERK   |  3300.00 | [null]       |        121 |            50 | Shipping
         130 | Mozhe      | Atkinson   | MATKINSO | 650.124.6234       | 1997-10-30 00:00:00 | ST_CLERK   |  2800.00 | [null]       |        121 |            50 | Shipping
         131 | James      | Marlow     | JAMRLOW  | 650.124.7234       | 1997-02-16 00:00:00 | ST_CLERK   |  2500.00 | [null]       |        121 |            50 | Shipping
         132 | TJ         | Olson      | TJOLSON  | 650.124.8234       | 1999-04-10 00:00:00 | ST_CLERK   |  2100.00 | [null]       |        121 |            50 | Shipping
         133 | Jason      | Mallin     | JMALLIN  | 650.127.1934       | 1996-06-14 00:00:00 | ST_CLERK   |  3300.00 | [null]       |        122 |            50 | Shipping
         134 | Michael    | Rogers     | MROGERS  | 650.127.1834       | 1998-08-26 00:00:00 | ST_CLERK   |  2900.00 | [null]       |        122 |            50 | Shipping
         135 | Ki         | Gee        | KGEE     | 650.127.1734       | 1999-12-12 00:00:00 | ST_CLERK   |  2400.00 | [null]       |        122 |            50 | Shipping
         136 | Hazel      | Philtanker | HPHILTAN | 650.127.1634       | 2000-02-06 00:00:00 | ST_CLERK   |  2200.00 | [null]       |        122 |            50 | Shipping
         137 | Renske     | Ladwig     | RLADWIG  | 650.121.1234       | 1995-07-14 00:00:00 | ST_CLERK   |  3600.00 | [null]       |        123 |            50 | Shipping
         138 | Stephen    | Stiles     | SSTILES  | 650.121.2034       | 1997-10-26 00:00:00 | ST_CLERK   |  3200.00 | [null]       |        123 |            50 | Shipping
         139 | John       | Seo        | JSEO     | 650.121.2019       | 1998-02-12 00:00:00 | ST_CLERK   |  2700.00 | [null]       |        123 |            50 | Shipping
         140 | Joshua     | Patel      | JPATEL   | 650.121.1834       | 1998-04-06 00:00:00 | ST_CLERK   |  2500.00 | [null]       |        123 |            50 | Shipping
         141 | Trenna     | Rajs       | TRAJS    | 650.121.8009       | 1995-10-17 00:00:00 | ST_CLERK   |  3500.00 | [null]       |        124 |            50 | Shipping
         142 | Curtis     | Davies     | CDAVIES  | 650.121.2994       | 1997-01-29 00:00:00 | ST_CLERK   |  3100.00 | [null]       |        124 |            50 | Shipping
         143 | Randall    | Matos      | RMATOS   | 650.121.2874       | 1998-03-15 00:00:00 | ST_CLERK   |  2600.00 | [null]       |        124 |            50 | Shipping
         144 | Peter      | Vargas     | PVARGAS  | 650.121.2004       | 1998-07-09 00:00:00 | ST_CLERK   |  2500.00 | [null]       |        124 |            50 | Shipping
         145 | John       | Russell    | JRUSSEL  | 011.44.1344.429268 | 1996-10-01 00:00:00 | SA_MAN     | 14000.00 |         0.40 |        100 |            80 | Sales
         146 | Karen      | Partners   | KPARTNER | 011.44.1344.467268 | 1997-01-05 00:00:00 | SA_MAN     | 13500.00 |         0.30 |        100 |            80 | Sales
         147 | Alberto    | Errazuriz  | AERRAZUR | 011.44.1344.429278 | 1997-03-10 00:00:00 | SA_MAN     | 12000.00 |         0.30 |        100 |            80 | Sales
         148 | Gerald     | Cambrault  | GCAMBRAU | 011.44.1344.619268 | 1999-10-15 00:00:00 | SA_MAN     | 11000.00 |         0.30 |        100 |            80 | Sales
         149 | Eleni      | Zlotkey    | EZLOTKEY | 011.44.1344.429018 | 2000-01-29 00:00:00 | SA_MAN     | 10500.00 |         0.20 |        100 |            80 | Sales
         150 | Peter      | Tucker     | PTUCKER  | 011.44.1344.129268 | 1997-01-30 00:00:00 | SA_REP     | 10000.00 |         0.30 |        145 |            80 | Sales
         151 | David      | Bernstein  | DBERNSTE | 011.44.1344.345268 | 1997-03-24 00:00:00 | SA_REP     |  9500.00 |         0.25 |        145 |            80 | Sales
```

Latihan No. 25

Judul
FULL [OUTER] JOIN: hanya baris yang tidak punya pasangan di salah satu sisi

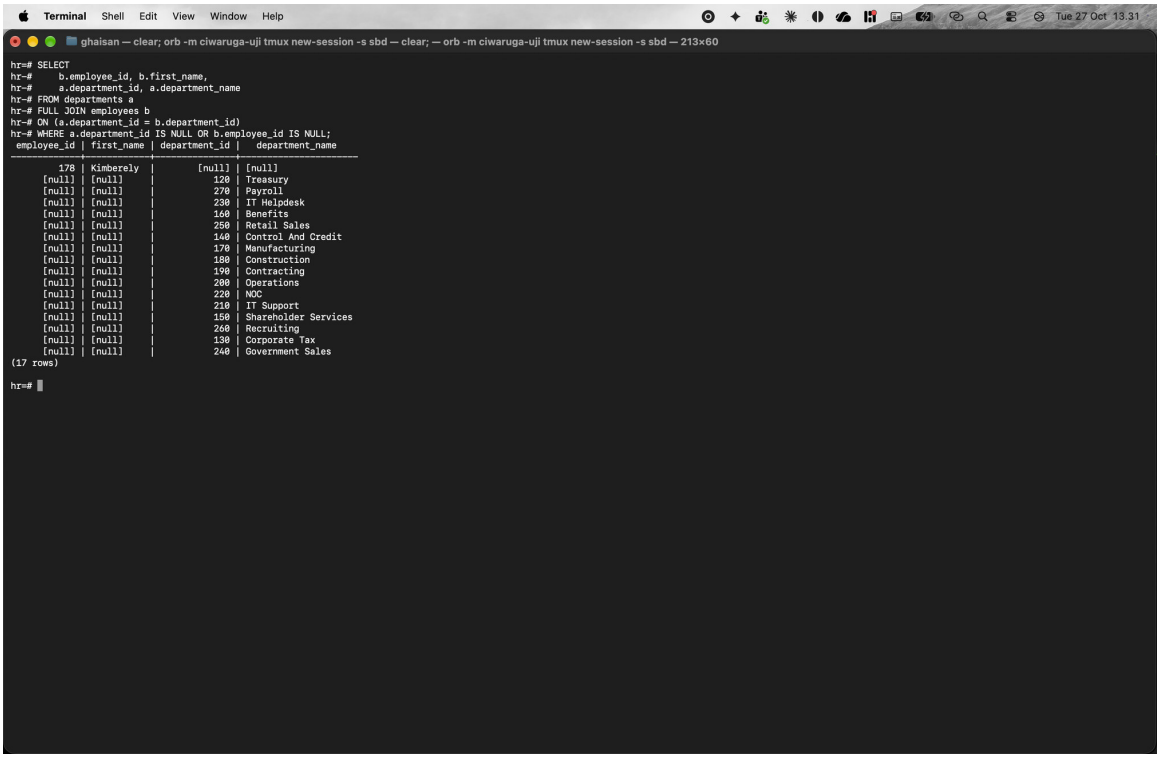
Syntax

```
SELECT
    b.employee_id, b.first_name,
    a.department_id, a.department_name
FROM departments a
FULL JOIN employees b
ON (a.department_id = b.department_id)
WHERE a.department_id IS NULL OR b.employee_id IS NULL;
```

Keterangan

Dengan WHERE a.department_id IS NULL OR b.employee_id IS NULL tersisa 17 baris yang tidak punya pasangan: 1 pegawai tanpa departemen (178 Kimberely) dan 16 departemen tanpa pegawai.

Hasil Eksekusi (Screenshot)



Latihan No. 26

Judul
JOIN tiga tabel: employees, departments, dan jobs

Syntax

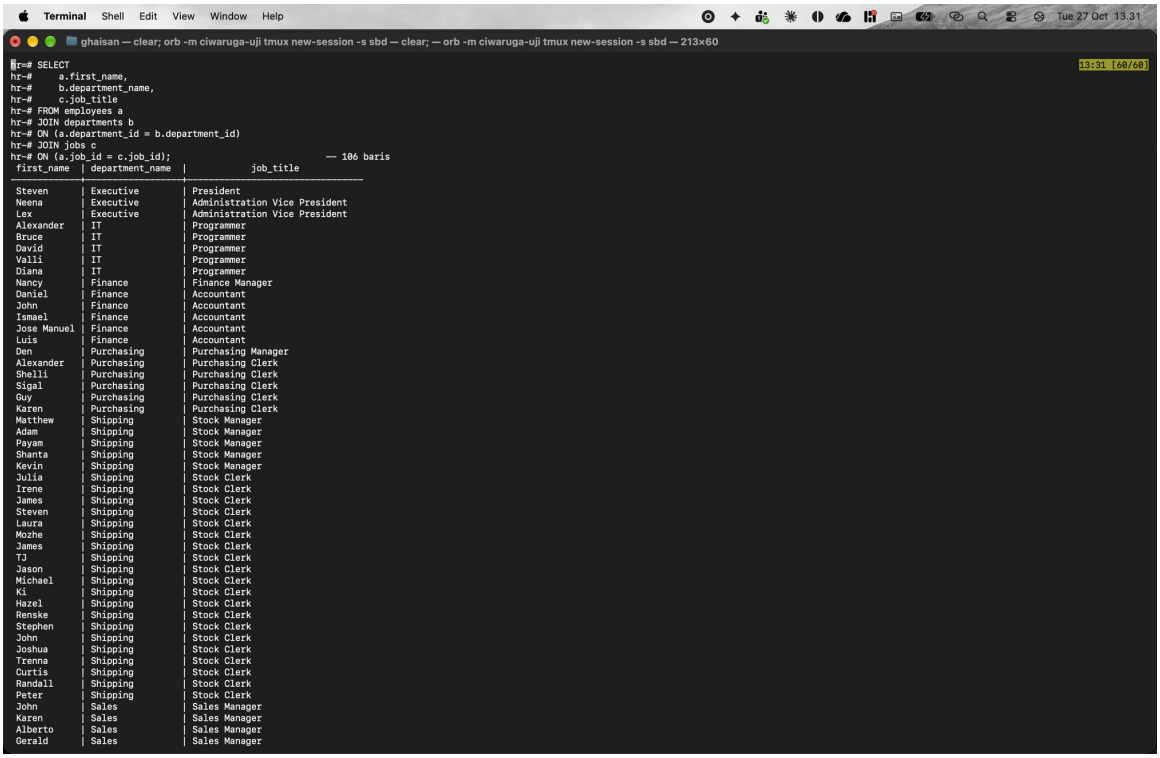
```
SELECT
    a.first_name,
    b.department_name,
    c.job_title
FROM employees a
JOIN departments b
ON (a.department_id = b.department_id)
JOIN jobs c
ON (a.job_id = c.job_id);                              -- 106 baris
```

Keterangan

Tiga tabel butuh dua kondisi join (n tabel butuh n - 1 kondisi). employees digabung dengan departments lewat department_id, lalu hasilnya digabung dengan jobs lewat job_id untuk mengambil job_title. Hasilnya 106 baris karena join pertama INNER JOIN, jadi Kimberely Grant tidak muncul.

Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
hr-# SELECT
hr-#     a.first_name,
hr-#     b.department_name,
hr-#     c.job_title
hr-# FROM employees a
hr-# JOIN departments b
hr-# ON (a.department_id = b.department_id)
hr-# JOIN jobs c
hr-# ON (a.job_id = c.job_id);                          -- 106 baris
 first_name  | department_name  |          job_title
-------------+------------------+-------------------------------
 Steven      | Executive        | President
 Neena       | Executive        | Administration Vice President
 Lex         | Executive        | Administration Vice President
 Alexander   | IT               | Programmer
 Bruce       | IT               | Programmer
 David       | IT               | Programmer
 Valli       | IT               | Programmer
 Diana       | IT               | Programmer
 Nancy       | Finance          | Finance Manager
 Daniel      | Finance          | Accountant
 John        | Finance          | Accountant
 Ismael      | Finance          | Accountant
 Jose Manuel | Finance          | Accountant
 Luis        | Finance          | Accountant
 Den         | Purchasing       | Purchasing Manager
 Alexander   | Purchasing       | Purchasing Clerk
 Shelli      | Purchasing       | Purchasing Clerk
 Sigal       | Purchasing       | Purchasing Clerk
 Guy         | Purchasing       | Purchasing Clerk
 Karen       | Purchasing       | Purchasing Clerk
 Matthew     | Shipping         | Stock Manager
 Adam        | Shipping         | Stock Manager
 Payam       | Shipping         | Stock Manager
 Shanta      | Shipping         | Stock Manager
 Kevin       | Shipping         | Stock Manager
 Julia       | Shipping         | Stock Clerk
 Irene       | Shipping         | Stock Clerk
 James       | Shipping         | Stock Clerk
 Steven      | Shipping         | Stock Clerk
 Laura       | Shipping         | Stock Clerk
 Mozhe       | Shipping         | Stock Clerk
 James       | Shipping         | Stock Clerk
 TJ          | Shipping         | Stock Clerk
 Jason       | Shipping         | Stock Clerk
 Michael     | Shipping         | Stock Clerk
 Ki          | Shipping         | Stock Clerk
 Hazel       | Shipping         | Stock Clerk
 Renske      | Shipping         | Stock Clerk
 Stephen     | Shipping         | Stock Clerk
 John        | Shipping         | Stock Clerk
 Joshua      | Shipping         | Stock Clerk
 Trenna      | Shipping         | Stock Clerk
 Curtis      | Shipping         | Stock Clerk
 Randall     | Shipping         | Stock Clerk
 Peter       | Shipping         | Stock Clerk
 John        | Sales            | Sales Manager
 Karen       | Sales            | Sales Manager
 Alberto     | Sales            | Sales Manager
 Gerald      | Sales            | Sales Manager
```

Latihan No. 27

Judul
JOIN tiga tabel: join pertama diganti LEFT JOIN supaya semua 107 pegawai tampil

Syntax

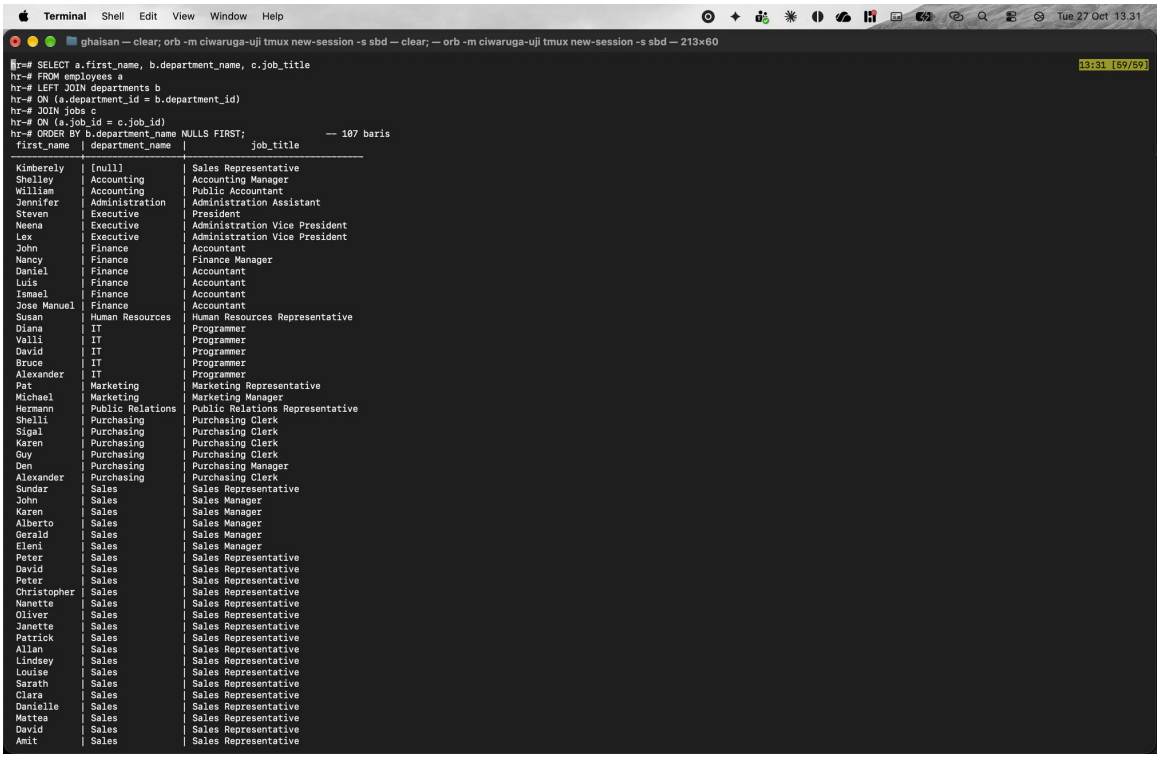
```
SELECT a.first_name, b.department_name, c.job_title
FROM employees a
LEFT JOIN departments b
ON (a.department_id = b.department_id)
JOIN jobs c
ON (a.job_id = c.job_id)
ORDER BY b.department_name NULLS FIRST;                -- 107 baris
```

Keterangan

Join pertama diganti LEFT JOIN, jadi semua 107 pegawai tampil. Kimberely Grant muncul dengan department_name [null]. ORDER BY ... NULLS FIRST ditambahkan supaya baris Kimberely ada di paling atas dan kelihatan di screenshot.

Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
Terminal  Shell  Edit  View  Window  Help                                   Tue 27 Oct 13.31
ghaisan — clear; orb -m ciwaruga-uji tmux new-session -s sbd — clear; — orb -m ciwaruga-uji tmux new-session -s sbd — 213×60
hr=# SELECT a.first_name, b.department_name, c.job_title                                13:31 [59/59]
hr-# FROM employees a
hr-# LEFT JOIN departments b
hr-# ON (a.department_id = b.department_id)
hr-# JOIN jobs c
hr-# ON (a.job_id = c.job_id)
hr-# ORDER BY b.department_name NULLS FIRST;              — 107 baris
 first_name  | department_name  |           job_title
-------------+------------------+--------------------------------
 Kimberely   | [null]           | Sales Representative
 Shelley     | Accounting       | Accounting Manager
 William     | Accounting       | Public Accountant
 Jennifer    | Administration   | Administration Assistant
 Steven      | Executive        | President
 Neena       | Executive        | Administration Vice President
 Lex         | Executive        | Administration Vice President
 John        | Finance          | Accountant
 Nancy       | Finance          | Finance Manager
 Daniel      | Finance          | Accountant
 Luis        | Finance          | Accountant
 Ismael      | Finance          | Accountant
 Jose Manuel | Finance          | Accountant
 Susan       | Human Resources  | Human Resources Representative
 Diana       | IT               | Programmer
 Valli       | IT               | Programmer
 David       | IT               | Programmer
 Bruce       | IT               | Programmer
 Alexander   | IT               | Programmer
 Pat         | Marketing        | Marketing Representative
 Michael     | Marketing        | Marketing Manager
 Hermann     | Public Relations | Public Relations Representative
 Shelli      | Purchasing       | Purchasing Clerk
 Sigal       | Purchasing       | Purchasing Clerk
 Karen       | Purchasing       | Purchasing Clerk
 Guy         | Purchasing       | Purchasing Clerk
 Den         | Purchasing       | Purchasing Manager
 Alexander   | Purchasing       | Purchasing Clerk
 Sundar      | Sales            | Sales Representative
 John        | Sales            | Sales Manager
 Karen       | Sales            | Sales Manager
 Alberto     | Sales            | Sales Manager
 Gerald      | Sales            | Sales Manager
 Eleni       | Sales            | Sales Manager
 Peter       | Sales            | Sales Representative
 David       | Sales            | Sales Representative
 Peter       | Sales            | Sales Representative
 Christopher | Sales            | Sales Representative
 Nanette     | Sales            | Sales Representative
 Oliver      | Sales            | Sales Representative
 Janette     | Sales            | Sales Representative
 Patrick     | Sales            | Sales Representative
 Allan       | Sales            | Sales Representative
 Lindsey     | Sales            | Sales Representative
 Louise      | Sales            | Sales Representative
 Sarath      | Sales            | Sales Representative
 Clara       | Sales            | Sales Representative
 Danielle    | Sales            | Sales Representative
 Mattea      | Sales            | Sales Representative
 David       | Sales            | Sales Representative
 Amit        | Sales            | Sales Representative
```

Latihan No. 28

Judul

SELF JOIN: setiap pegawai beserta atasannya

Syntax

```
SELECT
    e.employee_id,
    (e.first_name || ' ' || e.last_name) AS employee,
    e.job_id,
    e.manager_id,
    (m.first_name || ' ' || m.last_name) AS manager
FROM employees e               -- e = sebagai pegawai
LEFT JOIN employees m          -- m = sebagai atasan
ON (m.employee_id = e.manager_id)
ORDER BY manager NULLS FIRST;
```

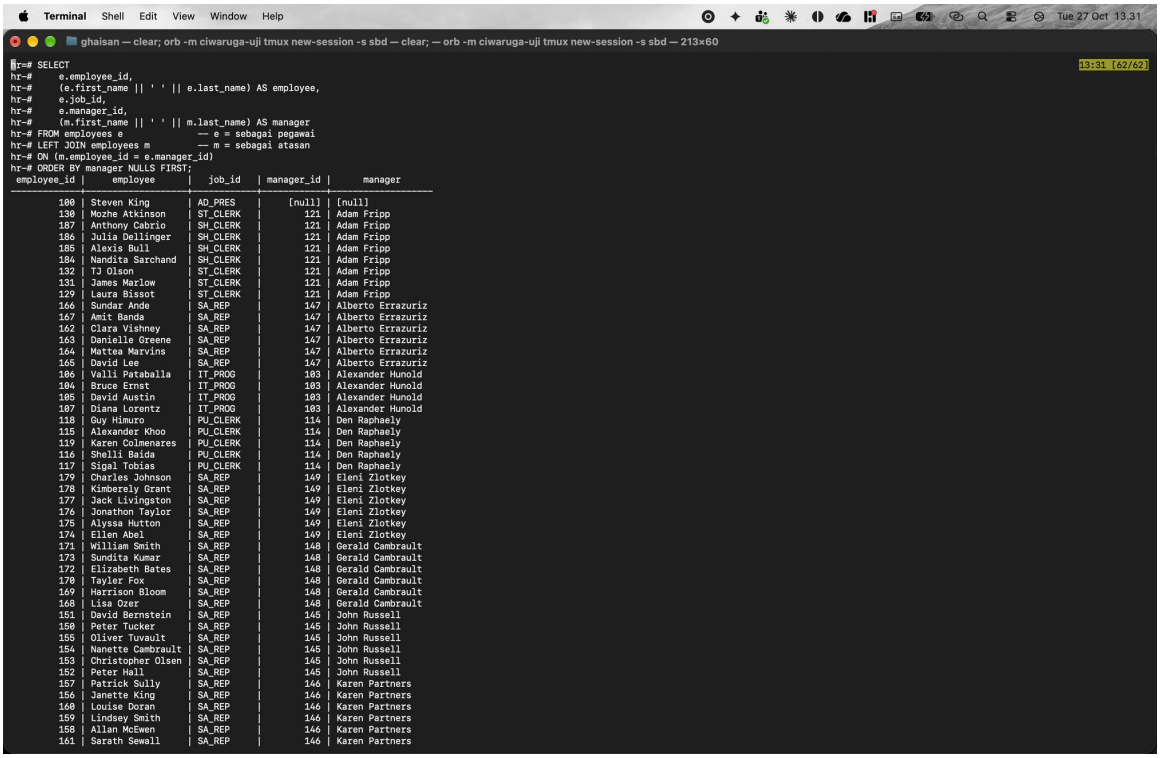
Keterangan

Tabel employees dipakai dua kali dengan alias berbeda: e sebagai pegawai dan m sebagai atasan, dihubungkan lewat m.employee_id = e.manager_id. LEFT JOIN dipakai supaya Steven King (AD_PRES) yang tidak punya atasan tetap tampil, totalnya 107 baris.

NULLS FIRST menaruh baris tanpa atasan di paling atas. Kolom manager Steven King [null] karena operator || menghasilkan NULL kalau salah satu operannya NULL.

Output lebih panjang dari satu layar, jadi screenshot menampilkan perintah dan bagian awal outputnya.

Hasil Eksekusi (Screenshot)



```
hr=# SELECT
hr-#     e.employee_id,
hr-#     (e.first_name || ' ' || e.last_name) AS employee,
hr-#     e.job_id,
hr-#     e.manager_id,
hr-#     (m.first_name || ' ' || m.last_name) AS manager
hr-# FROM employees e                -- e = sebagai pegawai
hr-# LEFT JOIN employees m           -- m = sebagai atasan
hr-# ON (m.employee_id = e.manager_id)
hr-# ORDER BY manager NULLS FIRST;
 employee_id |     employee      |   job_id   | manager_id |      manager
-------------+-------------------+------------+------------+-------------------
         100 | Steven King       | AD_PRES    |     [null] | [null]
         138 | Mozhe Atkinson    | ST_CLERK   |        121 | Adam Fripp
         187 | Anthony Cabrio    | SH_CLERK   |        121 | Adam Fripp
         186 | Julia Dellinger   | SH_CLERK   |        121 | Adam Fripp
         185 | Alexis Bull       | SH_CLERK   |        121 | Adam Fripp
         184 | Nandita Sarchand  | SH_CLERK   |        121 | Adam Fripp
         132 | TJ Olson          | ST_CLERK   |        121 | Adam Fripp
         131 | James Marlow      | ST_CLERK   |        121 | Adam Fripp
         129 | Laura Bissot      | ST_CLERK   |        121 | Adam Fripp
         166 | Sundar Ande       | SA_REP     |        147 | Alberto Errazuriz
         167 | Amit Banda        | SA_REP     |        147 | Alberto Errazuriz
         162 | Clara Vishney     | SA_REP     |        147 | Alberto Errazuriz
         163 | Danielle Greene   | SA_REP     |        147 | Alberto Errazuriz
         164 | Mattea Marvins    | SA_REP     |        147 | Alberto Errazuriz
         165 | David Lee         | SA_REP     |        147 | Alberto Errazuriz
         106 | Valli Pataballa   | IT_PROG    |        103 | Alexander Hunold
         104 | Bruce Ernst       | IT_PROG    |        103 | Alexander Hunold
         105 | David Austin      | IT_PROG    |        103 | Alexander Hunold
         107 | Diana Lorentz     | IT_PROG    |        103 | Alexander Hunold
         118 | Guy Himuro        | PU_CLERK   |        114 | Den Raphaely
         115 | Alexander Khoo    | PU_CLERK   |        114 | Den Raphaely
         119 | Karen Colmenares  | PU_CLERK   |        114 | Den Raphaely
         116 | Shelli Baida      | PU_CLERK   |        114 | Den Raphaely
         117 | Sigal Tobias      | PU_CLERK   |        114 | Den Raphaely
         179 | Charles Johnson   | SA_REP     |        149 | Eleni Zlotkey
         178 | Kimberely Grant   | SA_REP     |        149 | Eleni Zlotkey
         177 | Jack Livingston   | SA_REP     |        149 | Eleni Zlotkey
         176 | Jonathon Taylor   | SA_REP     |        149 | Eleni Zlotkey
         175 | Alyssa Hutton     | SA_REP     |        149 | Eleni Zlotkey
         174 | Ellen Abel        | SA_REP     |        149 | Eleni Zlotkey
         171 | William Smith     | SA_REP     |        148 | Gerald Cambrault
         173 | Sundita Kumar     | SA_REP     |        148 | Gerald Cambrault
         172 | Elizabeth Bates   | SA_REP     |        148 | Gerald Cambrault
         170 | Tayler Fox        | SA_REP     |        148 | Gerald Cambrault
         169 | Harrison Bloom    | SA_REP     |        148 | Gerald Cambrault
         168 | Lisa Ozer         | SA_REP     |        148 | Gerald Cambrault
         151 | David Bernstein   | SA_REP     |        145 | John Russell
         150 | Peter Tucker      | SA_REP     |        145 | John Russell
         155 | Oliver Tuvault    | SA_REP     |        145 | John Russell
         154 | Nanette Cambrault | SA_REP     |        145 | John Russell
         153 | Christopher Olsen | SA_REP     |        145 | John Russell
         152 | Peter Hall        | SA_REP     |        145 | John Russell
         157 | Patrick Sully     | SA_REP     |        146 | Karen Partners
         156 | Janette King      | SA_REP     |        146 | Karen Partners
         160 | Louise Doran      | SA_REP     |        146 | Karen Partners
         159 | Lindsey Smith     | SA_REP     |        146 | Karen Partners
         158 | Allan McEwen      | SA_REP     |        146 | Karen Partners
         161 | Sarath Sewall     | SA_REP     |        146 | Karen Partners
```

Latihan No. 29

Judul

SELF JOIN: COALESCE untuk atasan yang NULL dan INNER JOIN sebagai pembanding

Syntax

```
-- saran slide: COALESCE + CONCAT
SELECT
    e.employee_id,
    CONCAT(e.first_name, ' ', e.last_name) AS employee,
    COALESCE(CONCAT(m.first_name, ' ', m.last_name), '-') AS manager
FROM employees e
LEFT JOIN employees m ON (m.employee_id = e.manager_id)
WHERE e.manager_id IS NULL;

-- COALESCE + operator ||
SELECT
    e.employee_id,
    e.first_name || ' ' || e.last_name AS employee,
    COALESCE(m.first_name || ' ' || m.last_name, '-') AS manager
FROM employees e
LEFT JOIN employees m ON (m.employee_id = e.manager_id)
WHERE e.manager_id IS NULL;

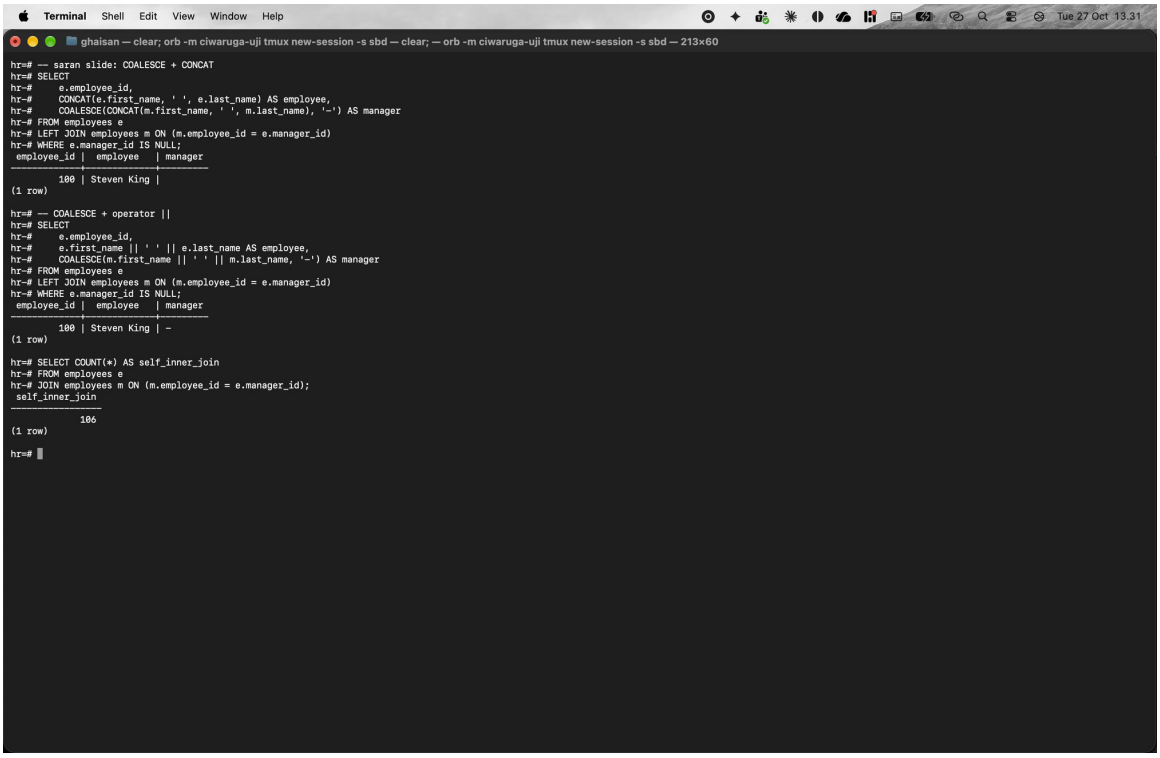
SELECT COUNT(*) AS self_inner_join
FROM employees e
JOIN employees m ON (m.employee_id = e.manager_id);
```

Keterangan

Slide menyarankan COALESCE(CONCAT(m.first_name, ' ', m.last_name), '-'). Setelah dicoba, di PostgreSQL hasilnya bukan '-' tapi teks berisi satu spasi. CONCAT menganggap NULL sebagai string kosong, jadi hasilnya ' ' (bukan NULL) dan COALESCE tidak mengganti apa-apa. Supaya '-' muncul, COALESCE dipasangkan dengan operator || yang menghasilkan NULL.

Query terakhir membuktikan self join dengan INNER JOIN hanya 106 baris karena Steven King tidak ikut.

Hasil Eksekusi (Screenshot)



Latihan No. 30

Judul
SELF JOIN: bawahan langsung pegawai nomor 121 dan siapa pegawai nomor 121

Syntax

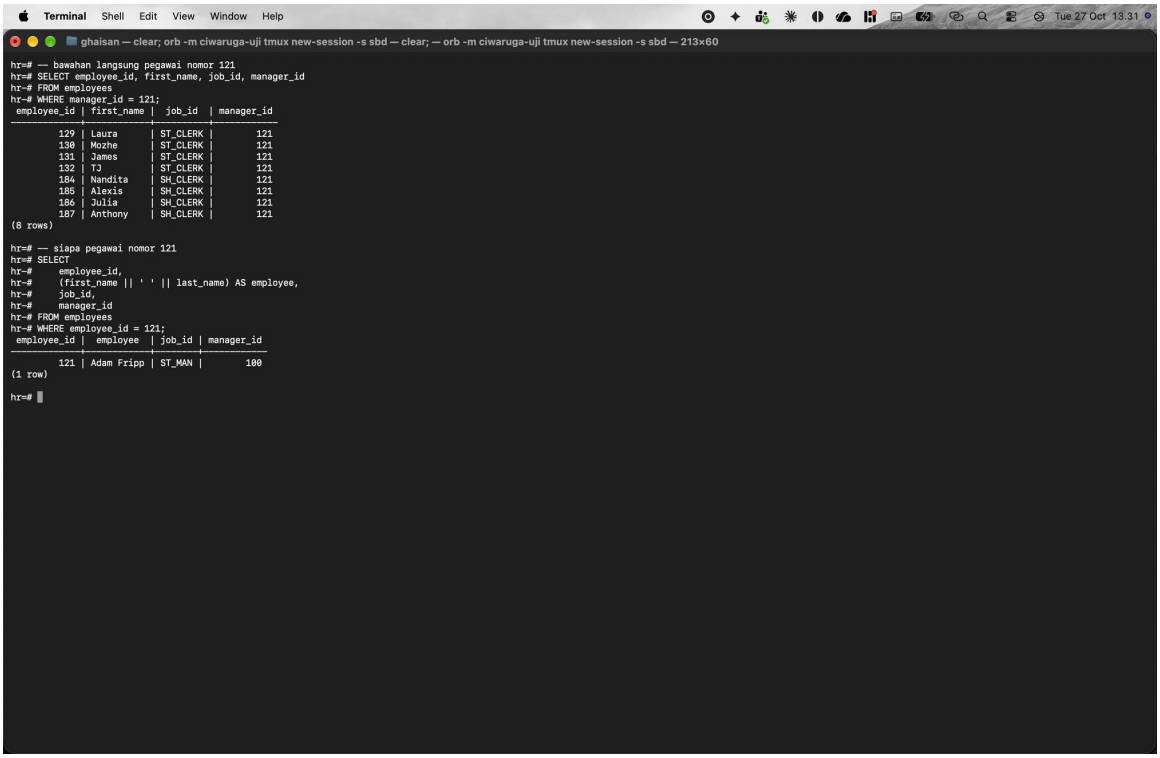
```
-- bawahan langsung pegawai nomor 121
SELECT employee_id, first_name, job_id, manager_id
FROM employees
WHERE manager_id = 121;

-- siapa pegawai nomor 121
SELECT
    employee_id,
    (first_name || ' ' || last_name) AS employee,
    job_id,
    manager_id
FROM employees
WHERE employee_id = 121;
```

Keterangan

Pegawai dengan manager_id 121 ada 8 orang (4 ST_CLERK dan 4 SH_CLERK). Pegawai nomor 121 sendiri adalah Adam Fripp (ST_MAN) dengan atasan nomor 100, Steven King.

Hasil Eksekusi (Screenshot)



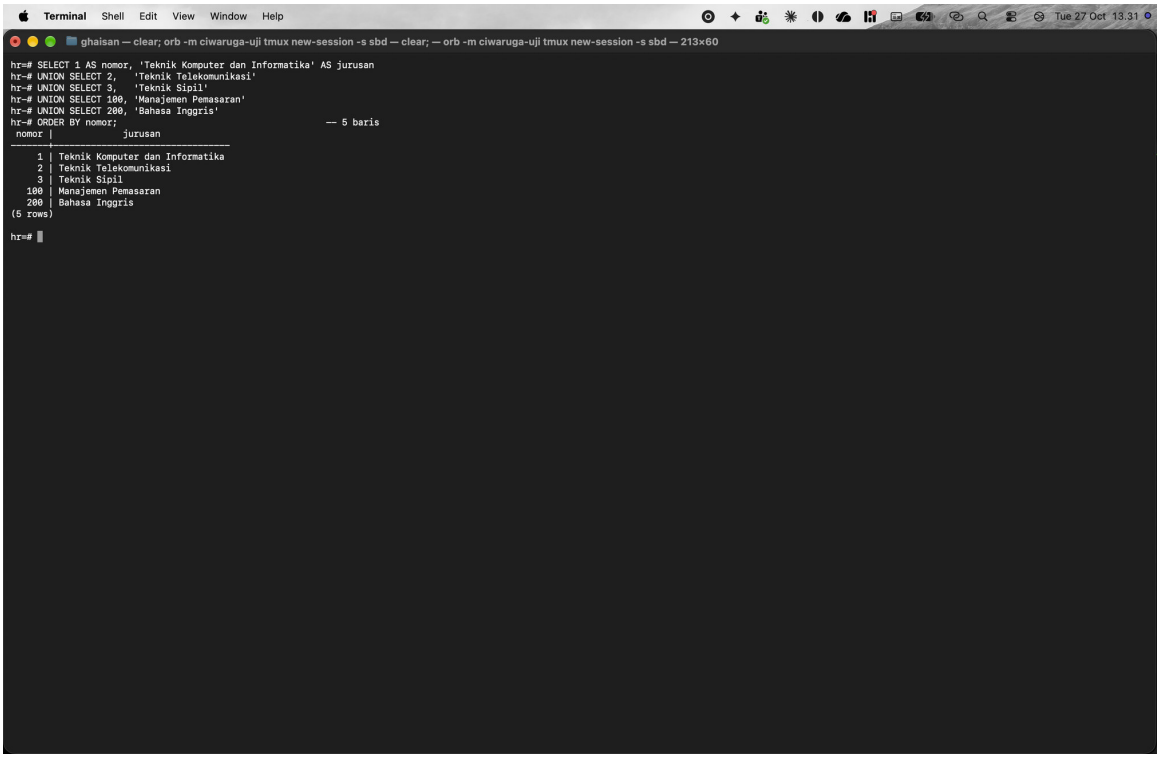
Latihan No. 31

Judul
Operator UNION: contoh sederhana dua daftar jurusan (A UNION B)
Syntax

`SELECT 1 AS nomor, 'Teknik Komputer dan Informatika' AS jurusan`
`UNION SELECT 2, 'Teknik Telekomunikasi'`
`UNION SELECT 3, 'Teknik Sipil'`
`UNION SELECT 100, 'Manajemen Pemasaran'`
`UNION SELECT 200, 'Bahasa Inggris'`
`ORDER BY nomor;  -- 5 baris`

Keterangan
UNION menumpuk hasil beberapa SELECT jadi satu. Contoh ini membangun A UNION B tanpa membuat tabel: 3 jurusan dari A dan 2 jurusan dari B jadi 5 baris. Nama kolom (nomor, jurusan) diambil dari SELECT pertama. Syaratnya jumlah kolom tiap SELECT sama, tipe datanya sesuai, dan urutan kolomnya sama.

Hasil Eksekusi (Screenshot)



Latihan No. 32

Judul

Operator UNION pada skema HR: query A dan query B riwayat pegawai Marketing

Syntax

```
-- Query A: pegawai yang saat ini di departemen Marketing
SELECT
    department_name,
    hire_date AS start_date,
    NULL::date AS end_date,
    last_name
FROM departments a
JOIN employees b ON (a.department_id = b.department_id)
WHERE a.department_name = 'Marketing';                -- 2 baris

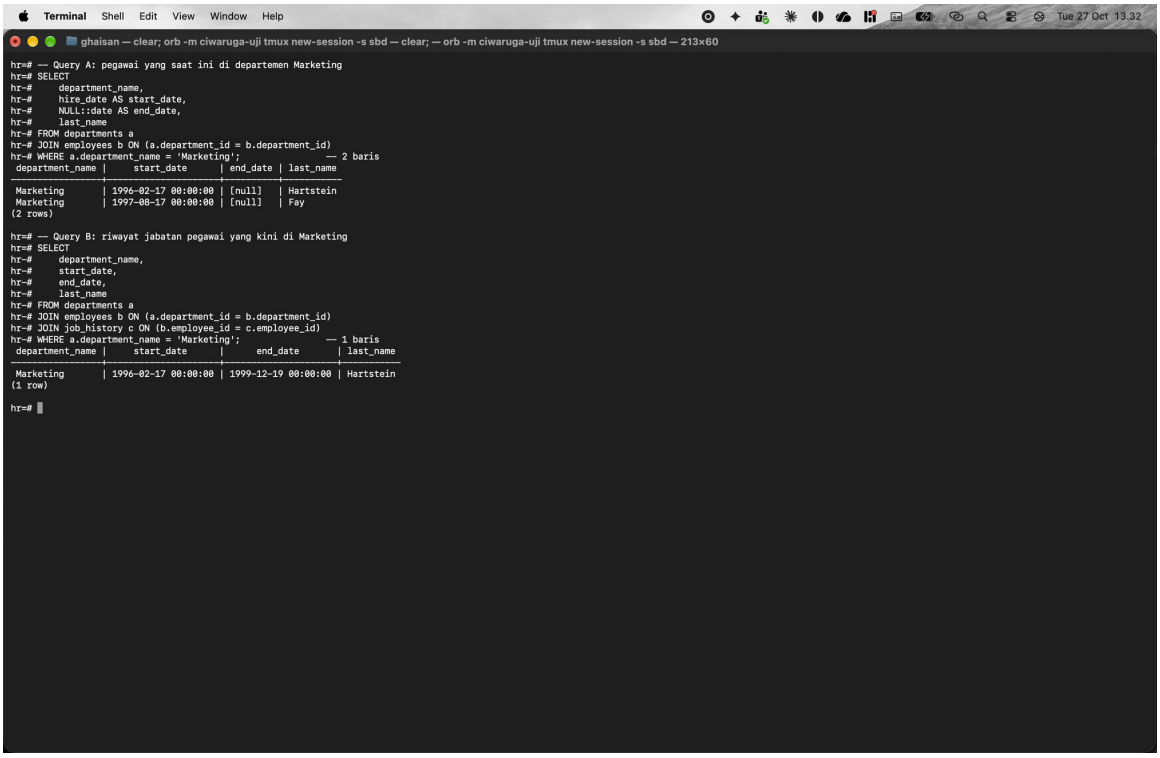
-- Query B: riwayat jabatan pegawai yang kini di Marketing
SELECT
    department_name,
    start_date,
    end_date,
    last_name
FROM departments a
JOIN employees b ON (a.department_id = b.department_id)
JOIN job_history c ON (b.employee_id = c.employee_id)
WHERE a.department_name = 'Marketing';                -- 1 baris
```

Keterangan

Query A mengambil pegawai yang sekarang di Marketing (Hartstein dan Fay) dengan hire_date sebagai start_date. end_date diisi NULL karena jabatan sekarang belum berakhir. Di Oracle cukup ditulis NULL, di PostgreSQL ditulis NULL::date supaya tipe kolomnya jelas.

Query B mengambil riwayat jabatan dari job_history, hasilnya 1 baris (Hartstein, 1996-02-17 sampai 1999-12-19).

Hasil Eksekusi (Screenshot)



Latihan No. 33

Judul

Operator UNION pada skema HR: query A UNION query B

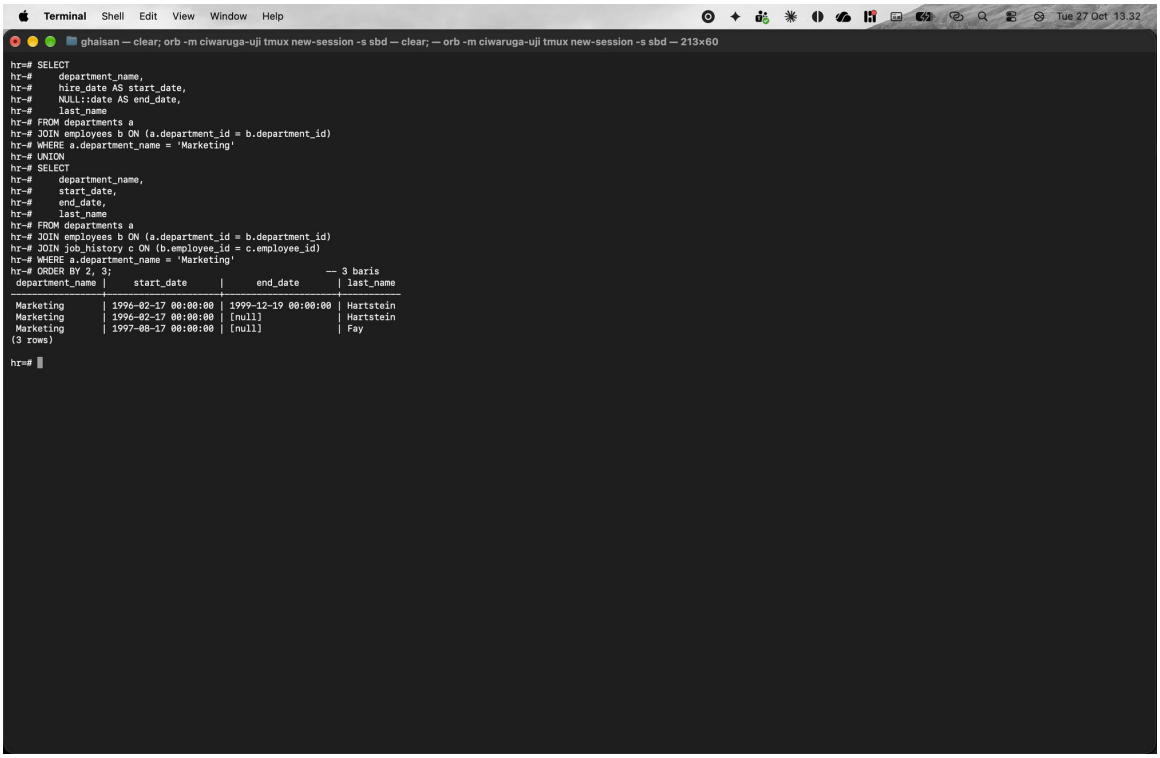
Syntax

```
SELECT
    department_name,
    hire_date AS start_date,
    NULL::date AS end_date,
    last_name
FROM departments a
JOIN employees b ON (a.department_id = b.department_id)
WHERE a.department_name = 'Marketing'
UNION
SELECT
    department_name,
    start_date,
    end_date,
    last_name
FROM departments a
JOIN employees b ON (a.department_id = b.department_id)
JOIN job_history c ON (b.employee_id = c.employee_id)
WHERE a.department_name = 'Marketing'
ORDER BY 2, 3;                                    -- 3 baris
```

Keterangan

Hasil gabungannya 3 baris: jabatan Hartstein sebelumnya (berakhir 1999-12-19), jabatan Hartstein sekarang, dan jabatan Fay sekarang. ORDER BY ditulis sekali di akhir dan hanya boleh memakai nama atau nomor kolom hasil. ORDER BY 2, 3 berarti urut berdasarkan start_date lalu end_date.

Hasil Eksekusi (Screenshot)



Latihan No. 34

Judul
UNION dan UNION ALL: membuang dan mempertahankan duplikat

Syntax

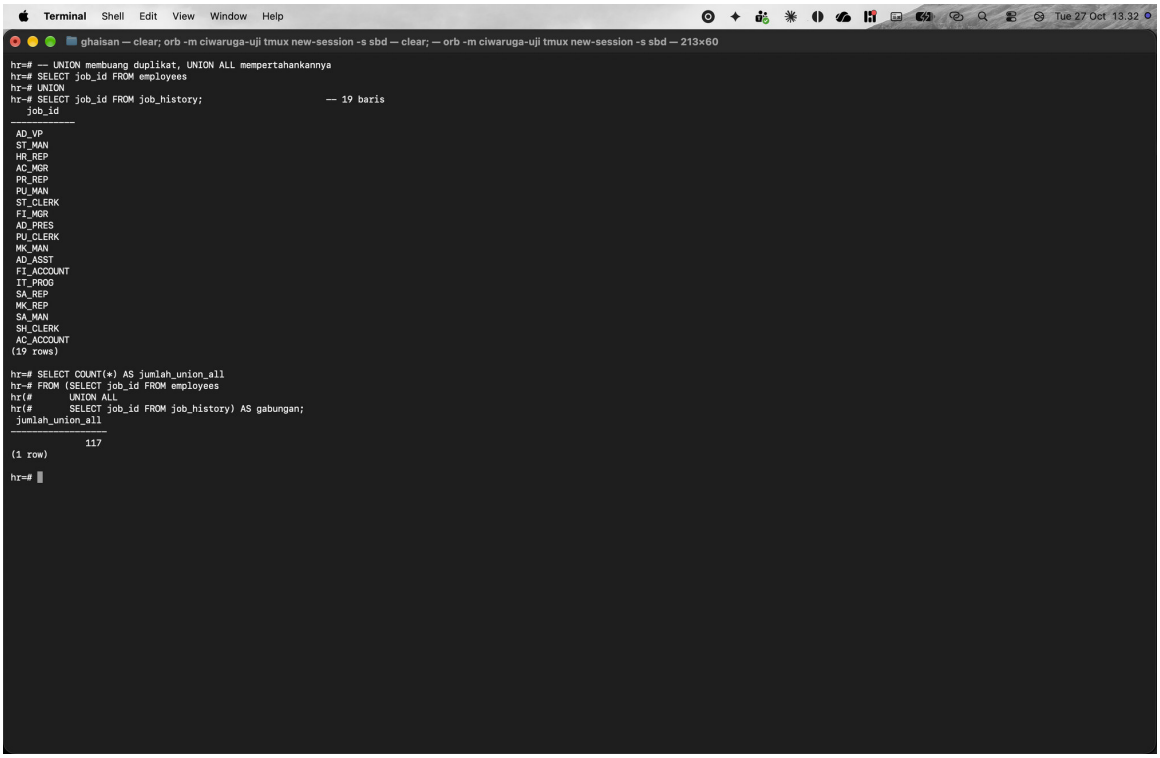
```
-- UNION membuang duplikat, UNION ALL mempertahankannya
SELECT job_id FROM employees
UNION
SELECT job_id FROM job_history;                          -- 19 baris

SELECT COUNT(*) AS jumlah_union_all
FROM (SELECT job_id FROM employees
      UNION ALL
      SELECT job_id FROM job_history) AS gabungan;
```

Keterangan

UNION membuang baris duplikat, jadi job_id dari employees dan job_history tinggal 19 nilai berbeda. UNION ALL mempertahankan semua baris: 107 + 10 = 117 baris (dihitung dengan COUNT supaya output tidak terlalu panjang).

Hasil Eksekusi (Screenshot)



Latihan No. 35

Judul
INTERSECT dan EXCEPT (pengganti MINUS di Oracle)

Syntax

```
-- INTERSECT: nilai yang ada di kedua query
SELECT department_id FROM employees
INTERSECT
SELECT department_id FROM departments;

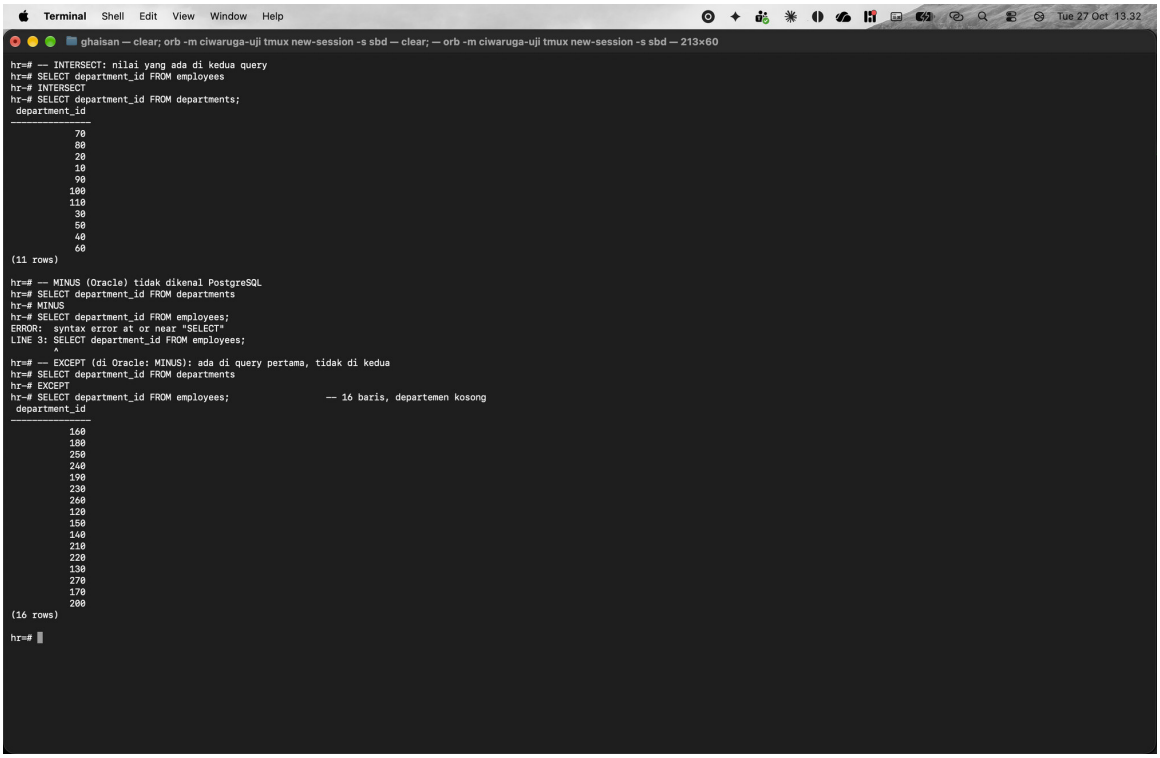
-- MINUS (Oracle) tidak dikenal PostgreSQL
SELECT department_id FROM departments
MINUS
SELECT department_id FROM employees;

-- EXCEPT (di Oracle: MINUS): ada di query pertama, tidak di kedua
SELECT department_id FROM departments
EXCEPT
SELECT department_id FROM employees;                    -- 16 baris, departemen
    kosong
```

Keterangan

INTERSECT mengambil nilai yang ada di kedua query: 11 department_id yang dipakai pegawai dan ada di departments (NULL milik Kimberely tidak ikut karena tidak ada di departments).
Oracle memakai MINUS untuk selisih himpunan. Di PostgreSQL MINUS dianggap syntax error, penggantinya EXCEPT. Hasil EXCEPT 16 department_id, yaitu departemen tanpa pegawai, sama dengan hasil LEFT JOIN ... IS NULL.

Hasil Eksekusi (Screenshot)



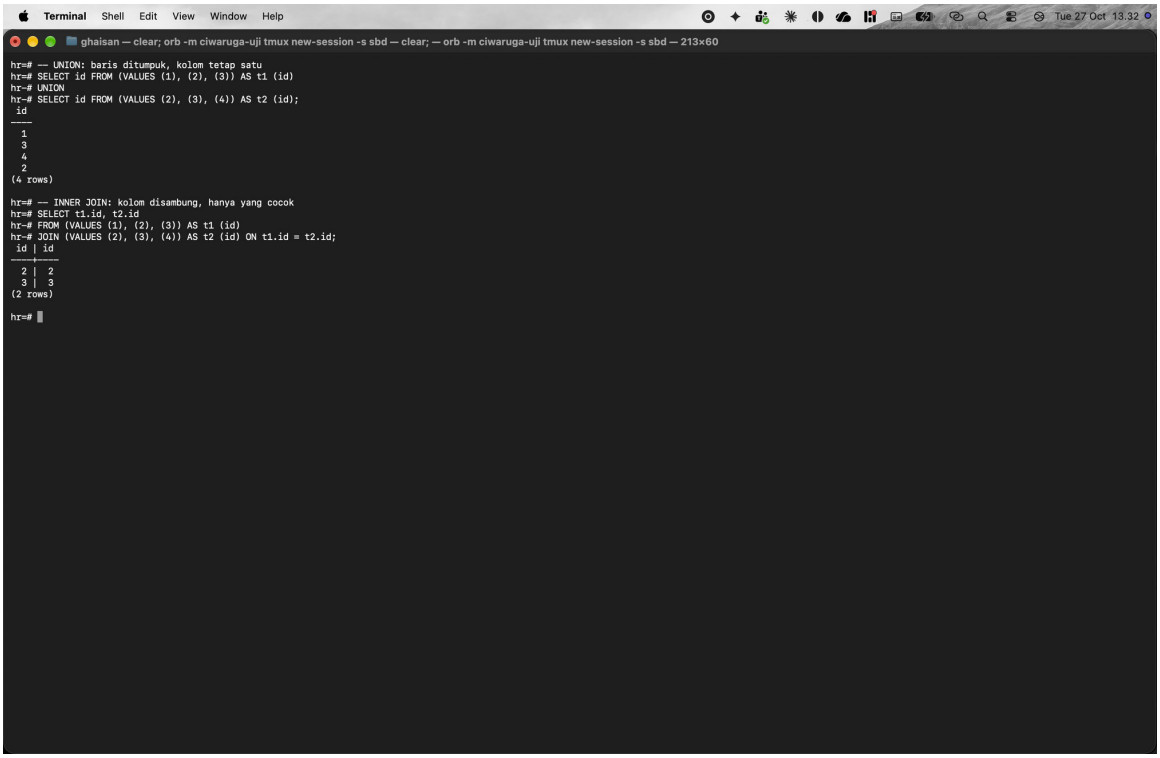
Latihan No. 36

Judul
UNION versus JOIN: baris ditumpuk dan kolom disambung
Syntax

`-- UNION: baris ditumpuk, kolom tetap satu`
`SELECT id FROM (VALUES (1), (2), (3)) AS t1 (id)`
`UNION`
`SELECT id FROM (VALUES (2), (3), (4)) AS t2 (id);`
 	
`-- INNER JOIN: kolom disambung, hanya yang cocok`
`SELECT t1.id, t2.id`
`FROM (VALUES (1), (2), (3)) AS t1 (id)`
`JOIN (VALUES (2), (3), (4)) AS t2 (id) ON t1.id = t2.id;`

Keterangan
UNION menumpuk hasil secara vertikal: kolomnya tetap satu (id) dan isinya 1, 2, 3, 4 tanpa duplikat. JOIN menyambung hasil secara horizontal: kolomnya jadi dua (t1.id dan t2.id) dan hanya pasangan yang cocok (2 dan 3) yang tampil. UNION menggabungkan berdasarkan posisi dan tipe kolom, JOIN berdasarkan kondisi ON. Urutan hasil UNION tidak tentu karena tidak ada ORDER BY.

Hasil Eksekusi (Screenshot)



Latihan No. 37

Judul

Join dengan fungsi bawaan dan WHERE: pegawai bergaji minimal 13000

Syntax

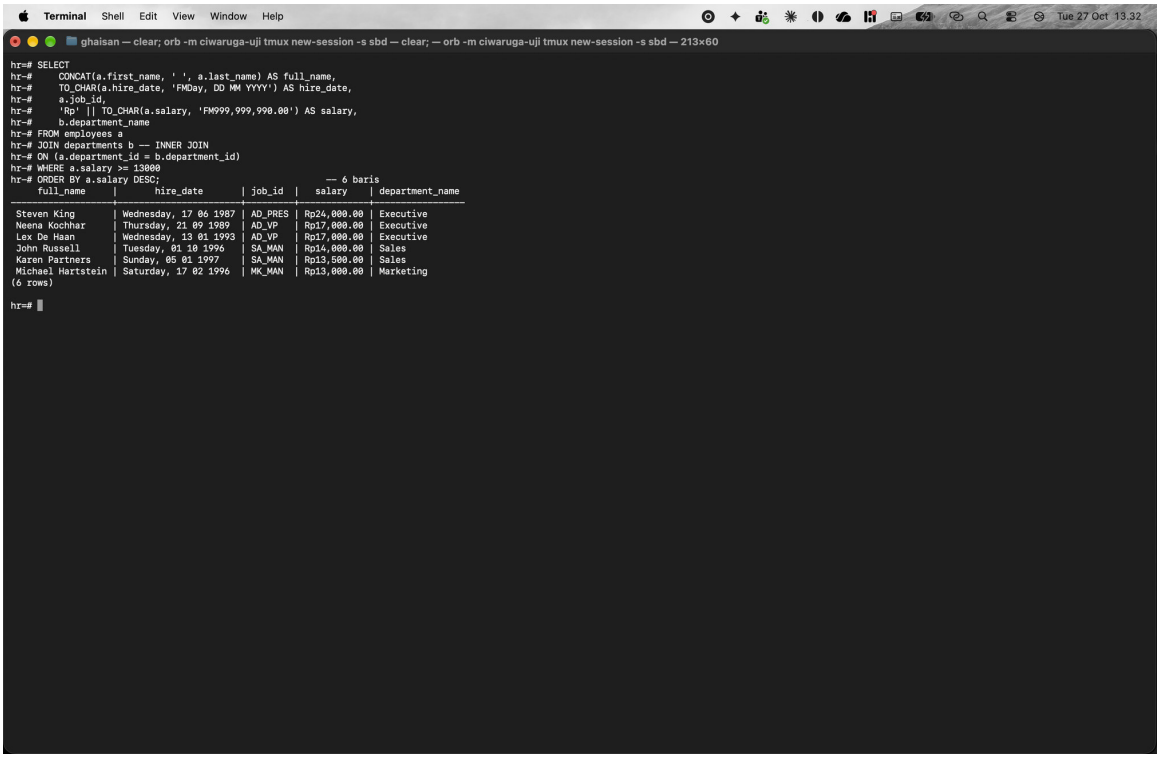
```
SELECT
    CONCAT(a.first_name, ' ', a.last_name) AS full_name,
    TO_CHAR(a.hire_date, 'FMDay, DD MM YYYY') AS hire_date,
    a.job_id,
    'Rp' || TO_CHAR(a.salary, 'FM999,999,990.00') AS salary,
    b.department_name
FROM employees a
JOIN departments b -- INNER JOIN
ON (a.department_id = b.department_id)
WHERE a.salary >= 13000
ORDER BY a.salary DESC;                                -- 6 baris
```

Keterangan

Versi Oracle di slide memakai CONCAT(CONCAT(a.first_name, ' '), a.last_name), TO_CHAR(hire_date, 'Day, DD MM YYYY'), dan TO_CHAR(a.salary, 'L999,999.99'). Di PostgreSQL CONCAT bisa menerima banyak argumen, format Day diberi FM supaya nama hari tidak ditambah spasi sampai 9 karakter, dan L diganti 'Rp' || TO_CHAR(..., 'FM999,999,990.00'). Hasilnya 6 pegawai bergaji minimal 13000.

Tahun hire_date berbeda dengan tabel hasil Oracle di slide (Steven King 1987, bukan 2003) karena data HR versi modb.pro memakai tanggal lama, sama seperti screenshot pgAdmin di slide.

Hasil Eksekusi (Screenshot)



Latihan No. 38

Judul
Join dengan fungsi bawaan dan WHERE: perangkap pengurutan kolom gaji yang sudah jadi teks

Syntax

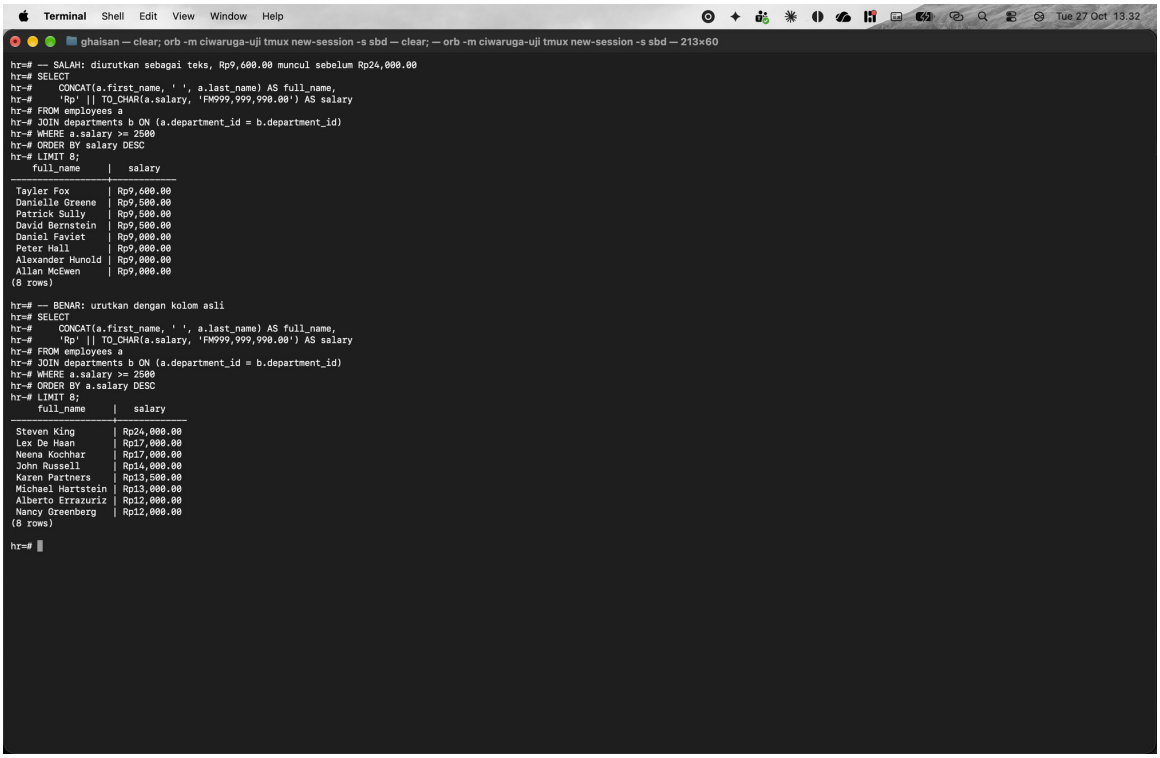
```
-- SALAH: diurutkan sebagai teks, Rp9,600.00 muncul sebelum Rp24,000.00
SELECT
    CONCAT(a.first_name, ' ', a.last_name) AS full_name,
    'Rp' || TO_CHAR(a.salary, 'FM999,999,990.00') AS salary
FROM employees a
JOIN departments b ON (a.department_id = b.department_id)
WHERE a.salary >= 2500
ORDER BY salary DESC
LIMIT 8;

-- BENAR: urutkan dengan kolom asli
SELECT
    CONCAT(a.first_name, ' ', a.last_name) AS full_name,
    'Rp' || TO_CHAR(a.salary, 'FM999,999,990.00') AS salary
FROM employees a
JOIN departments b ON (a.department_id = b.department_id)
WHERE a.salary >= 2500
ORDER BY a.salary DESC
LIMIT 8;
```

Keterangan

ORDER BY salary memakai kolom hasil salary yang sudah berupa teks, jadi diurutkan per karakter: 'Rp9,600.00' dianggap lebih besar dari 'Rp24,000.00' karena '9' lebih besar dari '2'. Hasilnya salah, Tayler Fox muncul paling atas. ORDER BY a.salary memakai kolom angka asli, jadi Steven King (24000) ada di urutan pertama.

Hasil Eksekusi (Screenshot)



Latihan No. 39

Judul
GROUP BY: jumlah pegawai dan total gaji setiap departemen

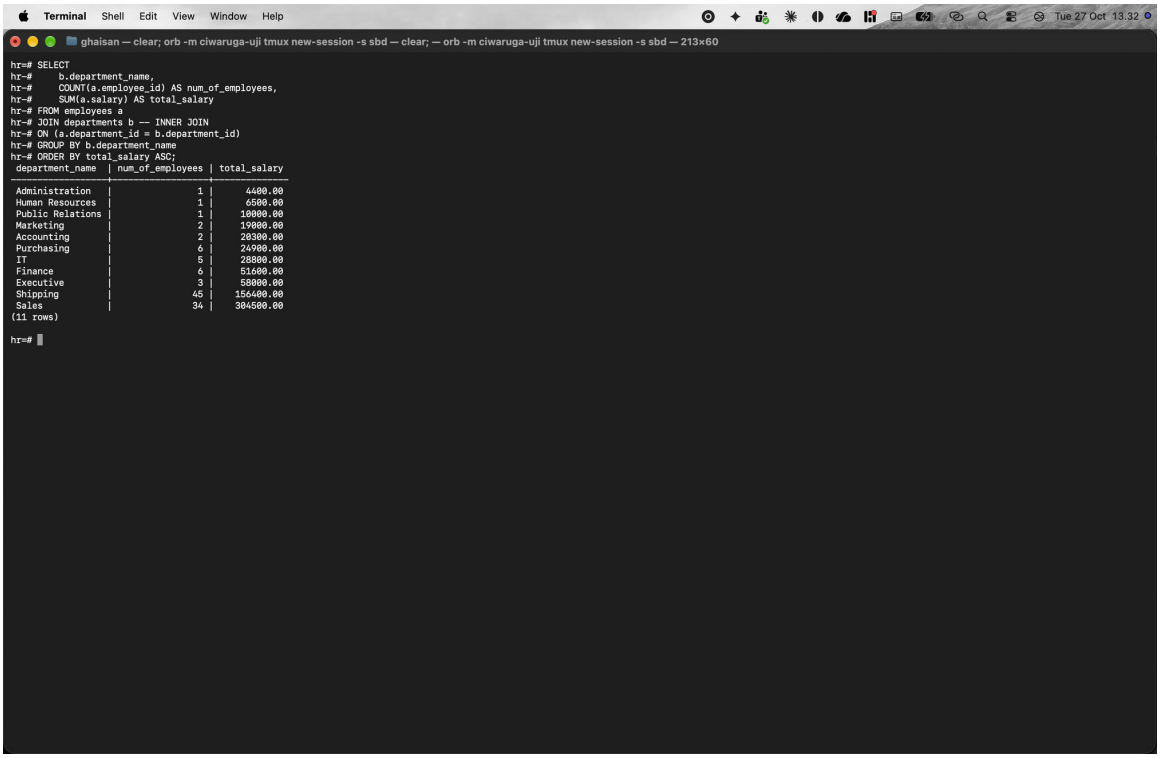
Syntax

```
SELECT
    b.department_name,
    COUNT(a.employee_id) AS num_of_employees,
    SUM(a.salary) AS total_salary
FROM employees a
JOIN departments b -- INNER JOIN
ON (a.department_id = b.department_id)
GROUP BY b.department_name
ORDER BY total_salary ASC;
```

Keterangan

GROUP BY b.department_name membuat satu baris untuk setiap departemen. COUNT menghitung jumlah pegawai dan SUM menjumlahkan gaji. Hasilnya 11 departemen dengan total pegawai 1+1+1+2+2+6+5+6+3+45+34 = 106, sama dengan hasil INNER JOIN. total_salary bertipe NUMERIC jadi tampil dengan dua desimal.

Hasil Eksekusi (Screenshot)



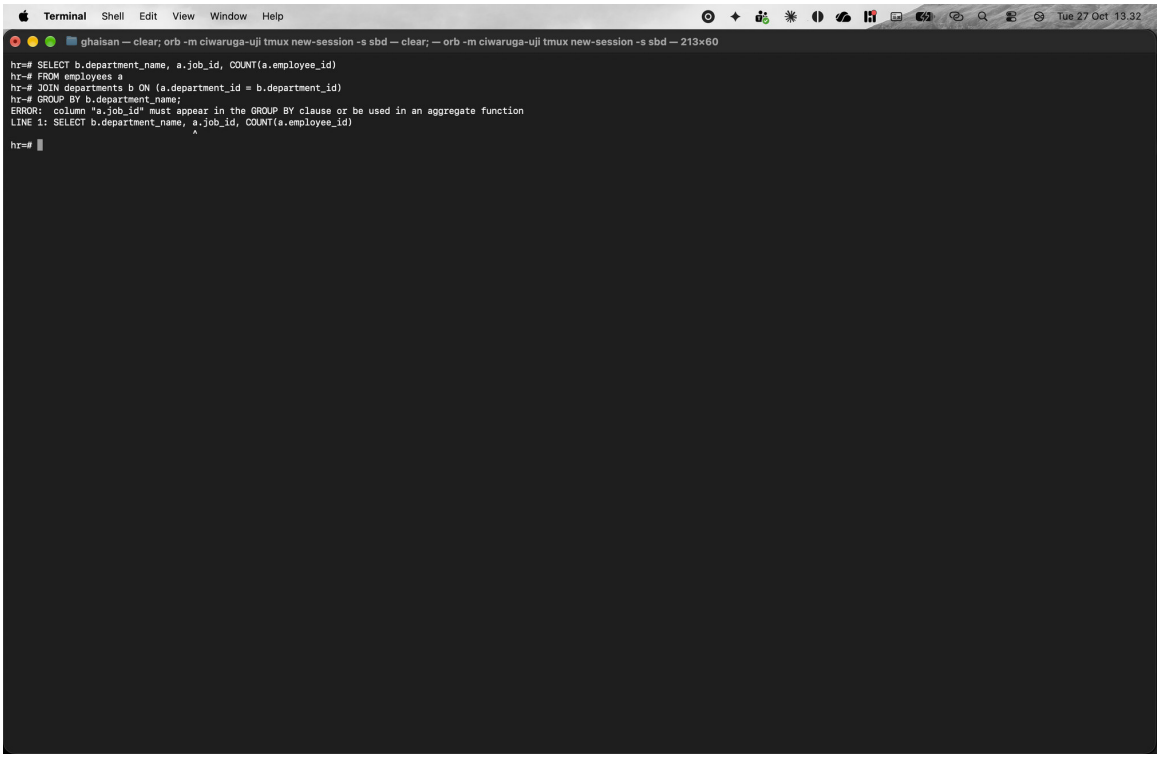
Latihan No. 40

Judul
GROUP BY: kolom yang bukan fungsi agregat wajib ada di GROUP BY
Syntax

`SELECT b.department_name, a.job_id, COUNT(a.employee_id)`
`FROM employees a`
`JOIN departments b ON (a.department_id = b.department_id)`
`GROUP BY b.department_name;`

Keterangan
a.job_id bukan fungsi agregat dan tidak ditulis di GROUP BY, jadi PostgreSQL menolak query dengan error column "a.job_id" must appear in the GROUP BY clause or be used in an aggregate function. Setiap kolom di SELECT yang bukan fungsi agregat harus ditulis juga di GROUP BY.

Hasil Eksekusi (Screenshot)



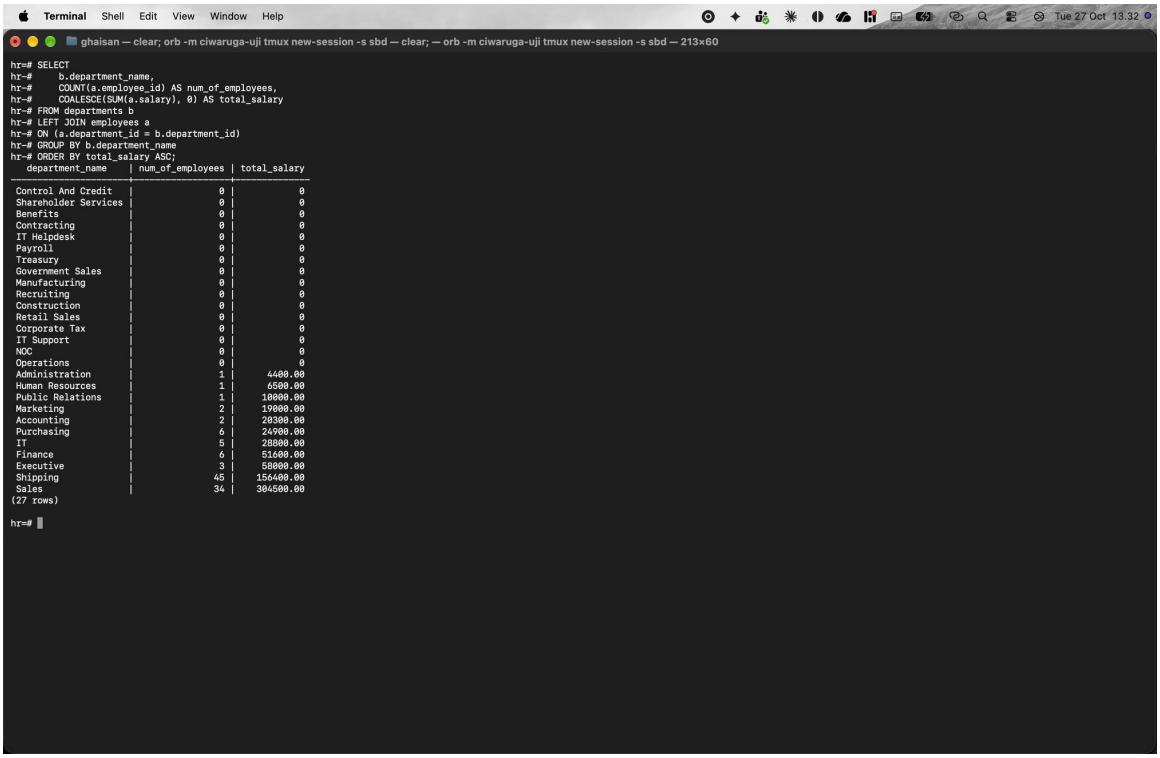
Latihan No. 41

Judul
GROUP BY dengan LEFT JOIN: departemen tanpa pegawai ikut tampil dengan jumlah 0
Syntax

`SELECT`
`  b.department_name,`
`  COUNT(a.employee_id) AS num_of_employees,`
`  COALESCE(SUM(a.salary), 0) AS total_salary`
`FROM departments b`
`LEFT JOIN employees a`
`ON (a.department_id = b.department_id)`
`GROUP BY b.department_name`
`ORDER BY total_salary ASC;`

Keterangan
departments dijadikan tabel kiri dengan LEFT JOIN, jadi 27 departemen tampil semua. Departemen tanpa pegawai punya num_of_employees 0 karena COUNT(a.employee_id) tidak menghitung NULL, dan total_salary 0 karena hasil SUM yang NULL diganti COALESCE.

Hasil Eksekusi (Screenshot)



Latihan No. 42

Judul
Ilustrasi ringkasan jenis join: tabel sementara tab1 dan tab2

Syntax

```
CREATE TEMP TABLE tab1 (numid INT);

INSERT INTO tab1 VALUES (12), (14), (10), (11);

CREATE TEMP TABLE tab2 (numid INT);

INSERT INTO tab2 VALUES (13), (15), (11), (12);

SELECT * FROM tab1 INNER JOIN tab2 ON tab1.numid = tab2.numid;

SELECT * FROM tab1 LEFT JOIN tab2 ON tab1.numid = tab2.numid;

SELECT * FROM tab1 RIGHT JOIN tab2 ON tab1.numid = tab2.numid;

SELECT * FROM tab1 FULL OUTER JOIN tab2 ON tab1.numid = tab2.numid;
```

Keterangan

Dua tabel sementara (TEMP TABLE) dipakai untuk merangkum jenis join. tab1 berisi 12, 14, 10, 11 dan tab2 berisi 13, 15, 11, 12. INNER JOIN hanya mengambil nilai yang ada di keduanya (11 dan 12). LEFT JOIN mengambil semua isi tab1 (10, 11, 12, 14), RIGHT JOIN semua isi tab2 (11, 12, 13, 15), dan FULL OUTER JOIN semua nilai dari kedua tabel, dengan [null] di sisi yang tidak punya pasangan.
Tabel sementara hilang sendiri ketika koneksi psql ditutup, jadi tidak perlu dihapus.

Hasil Eksekusi (Screenshot)

